

Network Modelling Assignment 2

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1 Task 1: Geometrically Weighted Terms in ERGMs

```
# stocnet suite
library(manynet)
library(netrics) # degree/density functions moved here in manynet 2.0.0
library(igraph)

# sna suite
library(network)
library(ergm)

# plotting and presentation
library(ggplot2)
library(patchwork)
library(texreg)
library(broom)
library(dplyr)
library(tidyr)

set.seed(42)
DATA <- "../Glasgow/"
```

1.1 Data Preparation

```
# Read adjacency matrices with manynet
friendship <- read_matrix(paste0(DATA, "f1.csv"), header = FALSE)

# Read node attributes
attributes <- read.csv(paste0(DATA, "demographic.csv"), header = TRUE)
alcohol <- read.csv(paste0(DATA, "alcohol.csv"), header = TRUE)
attributes$alcohol <- alcohol$Wave.1

# Basic descriptives using manynet / netrics
```

```

nvertices <- net_nodes(friendship)
nvertices

## [1] 129

nedges <- net_ties(friendship)
nedges

## [1] 448

density <- net_by_density(friendship) # moved to netrics in manynet 2.0.0
density

## [1] 0.0271

outdegree <- node_by_outdegree(friendship, normalized = FALSE)
mean(outdegree)

## [1] 3.472868

sd(outdegree)

## [1] 1.630172

indegree <- node_by_indegree(friendship, normalized = FALSE)
mean(indegree)

## [1] 3.472868

sd(indegree)

## [1] 2.267562

plot(outdegree) + plot(indegree)

# Add a name column so join_nodes can match by node index
attributes$name <- 1:nvertices

# Create network object for ERGM estimation (sna suite)
friendship_sna <- friendship |>
  mutate_nodes(name = 1:nvertices) |>
  join_nodes(attributes) |>
  as_network()

friendship_sna

## Network attributes:
##   vertices = 129
##   directed = TRUE
##   hyper = FALSE
##   loops = FALSE
##   multiple = FALSE
##   bipartite = FALSE
##   total edges= 448
##   missing edges= 0
##   non-missing edges= 448
##
## Vertex attribute names:
##   age alcohol gender vertex.names

```

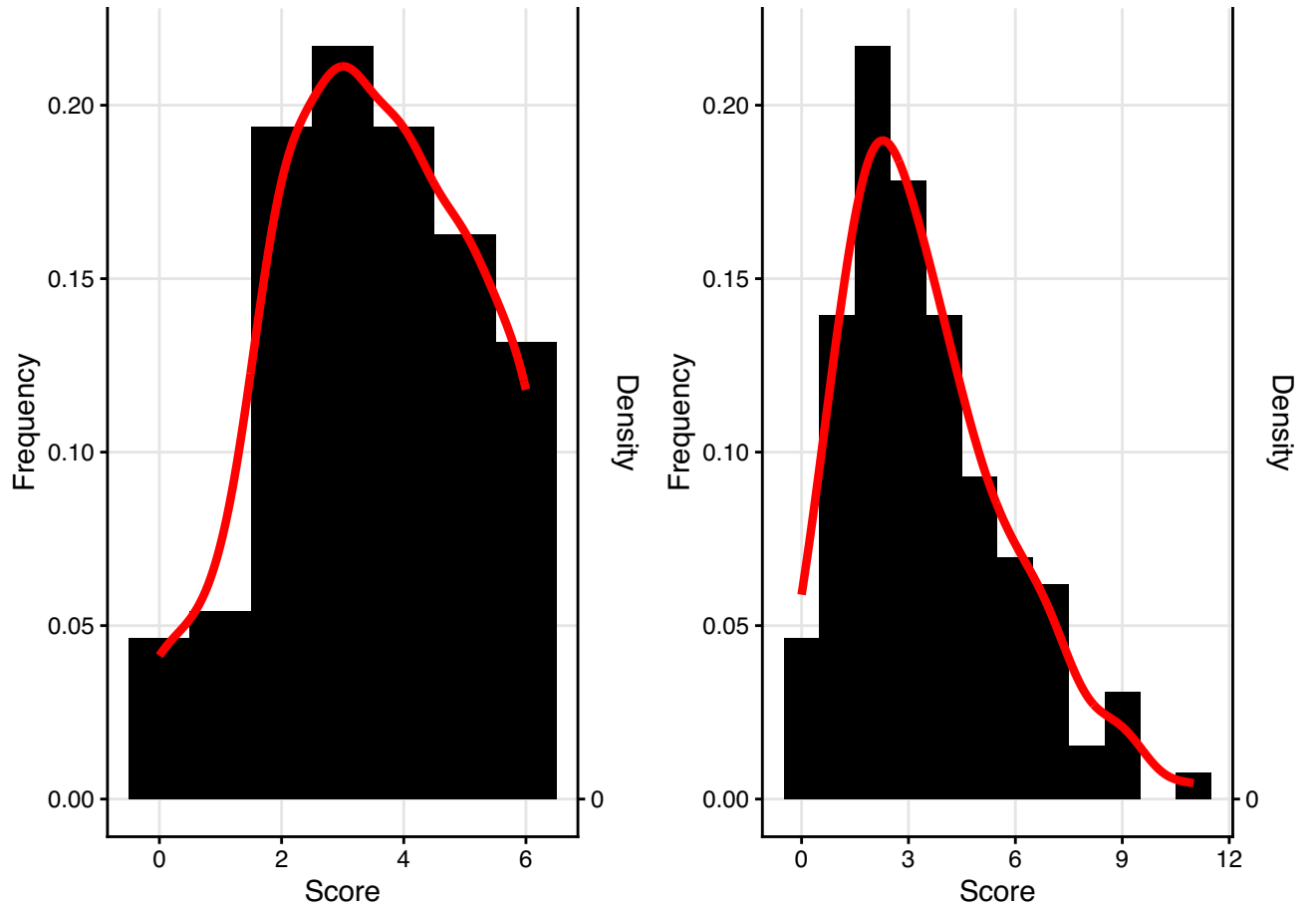


Figure 1: Out-degree and in-degree distributions of the Glasgow Wave 1 network.

```
##
## No edge attributes
```

1.2 Part (1): Change Statistic of gwodegree

1.2.1 (i) Analytical Derivation

The GWD network statistic is defined as:

$$s_{\text{GWD}}(x, \alpha) = e^\alpha \sum_{k=1}^n \left\{ 1 - (1 - e^{-\alpha})^k \right\} D_k^{\text{out}}(x)$$

where $D_k^{\text{out}}(x)$ is the number of nodes with exactly out-degree k .

Adding a directed edge from a sender node with current out-degree k :

- D_k^{out} decreases by 1 (the node leaves degree class k).
- D_{k+1}^{out} increases by 1 (the node enters degree class $k + 1$).

The change statistic is therefore:

$$\begin{aligned} \delta_{\text{GWD}}(k, \alpha) &= s_{\text{GWD}}(x + \text{edge}, \alpha) - s_{\text{GWD}}(x, \alpha) \\ &= e^\alpha \left[\left\{ 1 - (1 - e^{-\alpha})^{k+1} \right\} - \left\{ 1 - (1 - e^{-\alpha})^k \right\} \right] \\ &= e^\alpha \left[(1 - e^{-\alpha})^k - (1 - e^{-\alpha})^{k+1} \right] \\ &= e^\alpha (1 - e^{-\alpha})^k \underbrace{\left[1 - (1 - e^{-\alpha}) \right]}_{= e^{-\alpha}} = \boxed{(1 - e^{-\alpha})^k} \end{aligned}$$

1.2.2 (ii) Plots of the Change Statistic

```
delta_GWD <- function(k, alpha) (1 - exp(-alpha))^k

alpha_vals <- c(-1, -0.5, 0, 0.1, 0.2, 0.3, 0.5, log(2), 1, 2, 3)
alpha_labs <- c("-1", "-0.5", "0", "0.1", "0.2", "0.3",
               "0.5", "ln2", "1", "2", "3")

# Use the range of observed out-degrees
k_max <- max(outdegree)
k_range <- 0:k_max

df_plot <- do.call(rbind, lapply(seq_along(alpha_vals), function(i) {
  data.frame(
    k = k_range,
    delta = delta_GWD(k_range, alpha_vals[i]),
    alpha = factor(alpha_labs[i], levels = alpha_labs),
    group = ifelse(alpha_vals[i] < 0, "negative", "non-negative")
  )
}))
```

```
df_sub <- df_plot |> filter(alpha == "-1")
ggplot(df_sub, aes(x = k, y = delta)) +
  geom_hline(yintercept = 0, linetype = "dashed", colour = "grey60") +
  geom_line(colour = "steelblue", linewidth = 0.9) +
  geom_point(colour = "steelblue", size = 2.5) +
  labs(title = expression("Change statistic," ~ alpha == -1),
       subtitle = "Negative decay value",
       x = "Current out-degree k",
       y = expression(delta[GWD](k, alpha))) +
  theme_bw(base_size = 13)
```

Change statistic, $\alpha = -1$

Negative decay value

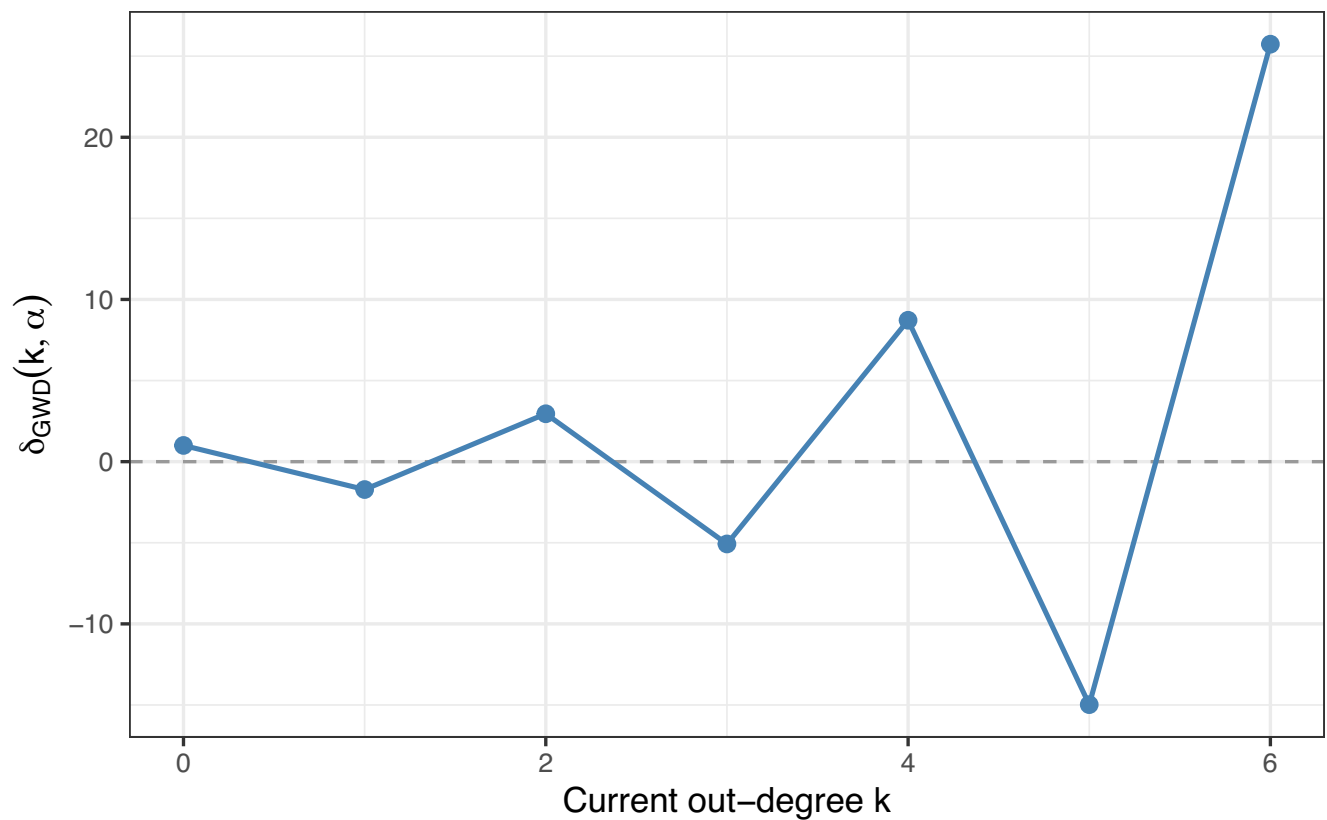


Figure 2: Change statistic for $\alpha = -1$.

```
df_sub <- df_plot |> filter(alpha == "-0.5")
ggplot(df_sub, aes(x = k, y = delta)) +
  geom_hline(yintercept = 0, linetype = "dashed", colour = "grey60") +
  geom_line(colour = "tomato", linewidth = 0.9) +
  geom_point(colour = "tomato", size = 2.5) +
  labs(title = expression("Change statistic," ~ alpha == -0.5),
       subtitle = "Negative decay value",
```

```
x = "Current out-degree k",
y = expression(delta[GWD](k, alpha)) +
theme_bw(base_size = 13)
```

Change statistic, $\alpha = -0.5$

Negative decay value

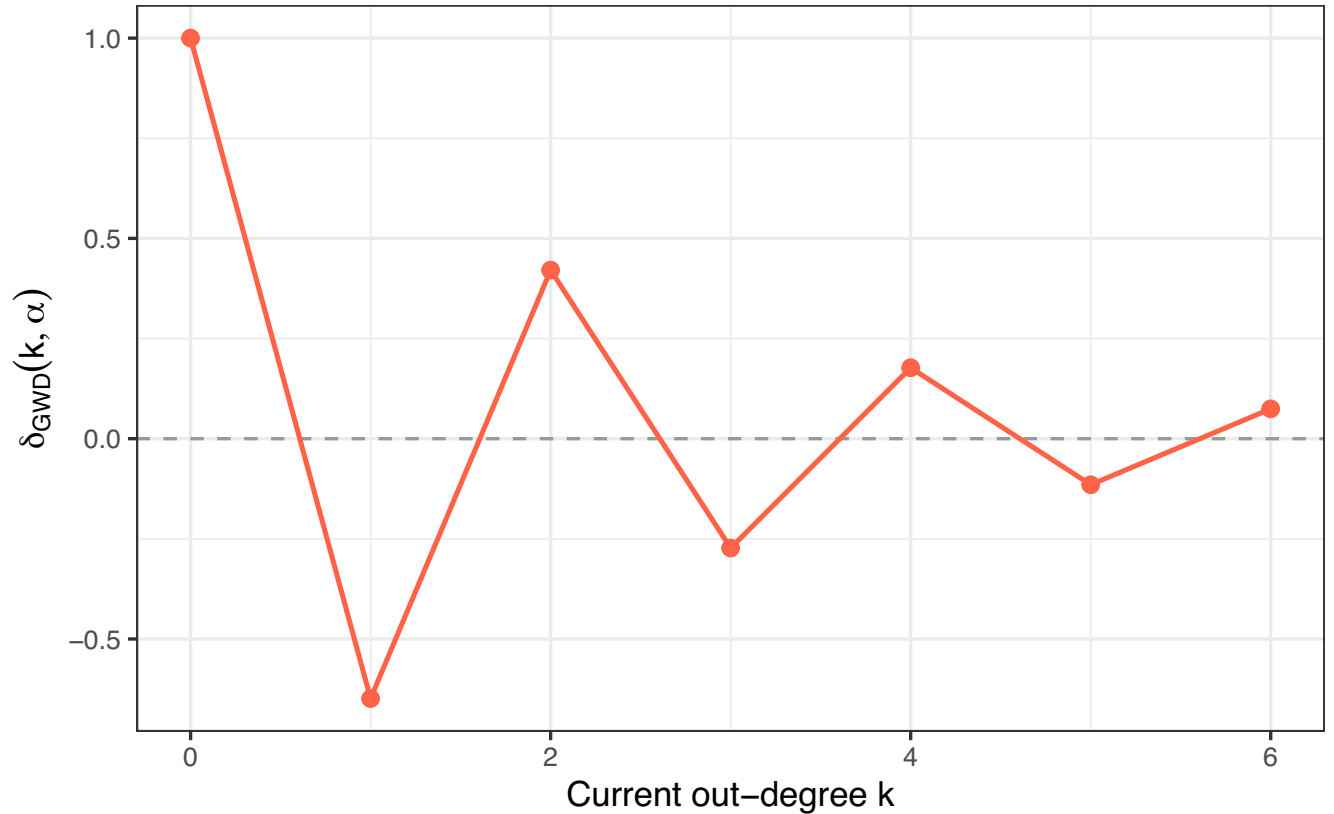


Figure 3: Change statistic for $\alpha = -0.5$.

1.2.2.1 Negative decay values (one figure per value)

```
df_nonneg <- df_plot |> filter(group == "non-negative")

ggplot(df_nonneg, aes(x = k, y = delta, colour = alpha, linetype = alpha)) +
  geom_line(linewidth = 0.9) +
  geom_point(size = 2.5) +
  scale_colour_brewer(palette = "Set1", name = expression(alpha)) +
  scale_linetype_discrete(name = expression(alpha)) +
  labs(title = "Change statistic - non-negative decay values",
       x = "Current out-degree k",
       y = expression(delta[GWD](k, alpha))) +
  theme_bw(base_size = 13)
```

1.2.2.2 Non-negative decay values (one combined figure)

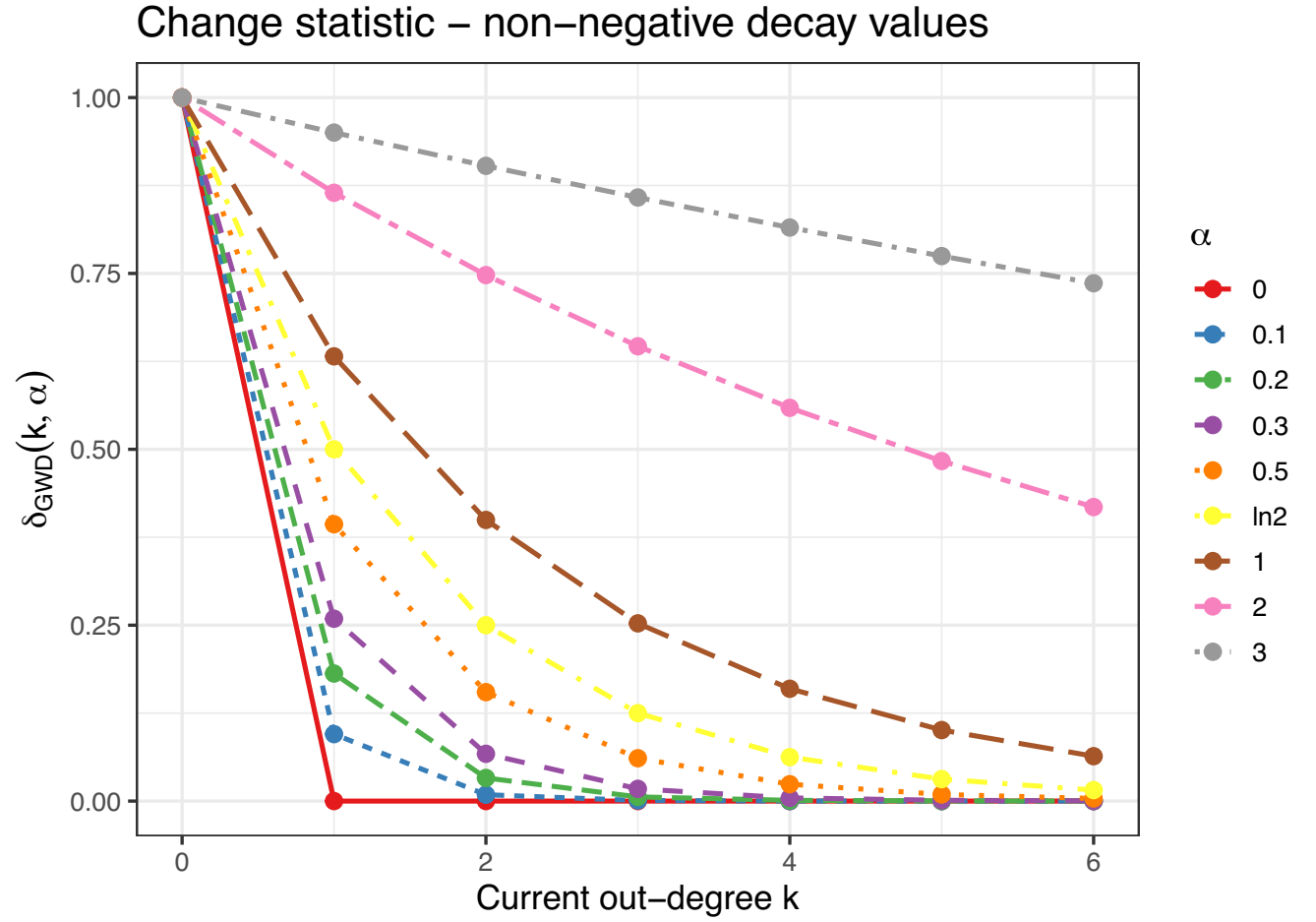


Figure 4: Change statistic for all non-negative decay values. Different colours and linetypes distinguish α values.

1.2.3 (iii) Main Takeaways

The change statistic $\delta_{\text{GWD}}(k, \alpha) = (1 - e^{-\alpha})^k$ reveals how the decay parameter α governs the marginal contribution of an additional out-edge to an already-popular sender node.

Non-negative α . When $\alpha > 0$ the base $(1 - e^{-\alpha}) \in (0, 1)$, so the change statistic is always positive and decreases *monotonically* with k . Adding an edge from a node with $k = 0$ always contributes the maximum value of 1, regardless of α , while adding from a high-degree node contributes progressively less. The rate of this decay is entirely determined by α :

- *Small α* (e.g. $\alpha = 0.1$): the base is close to 0, so δ drops almost immediately to near-zero after $k = 0$. Only ties sent by near-isolated nodes influence the statistic; hubs are practically invisible.
- *Large α* (e.g. $\alpha = 2$ or $\alpha = 3$): the base is close to 1, so δ remains substantial even at high k . Both low- and high-degree senders contribute almost equally, and the statistic is sensitive to the entire out-degree spectrum.
- $\alpha = \ln 2$: the base equals $\frac{1}{2}$, giving a clean geometric halving with each additional out-edge — a natural “neutral” reference.

When $\alpha = 0$ the change statistic is 0 for all $k > 0$: adding an edge from any already-active sender has no effect on the statistic whatsoever.

Negative α . The base $(1 - e^{-\alpha}) < 0$, so δ *alternates in sign* (positive for even k , negative for odd k) and grows in absolute magnitude. This has no natural interpretation in terms of a degree distribution; the statistic oscillates rather than capturing a smooth preferential-attachment or equalisation mechanism.

Summary. The decay parameter acts as a “bandwidth” on the out-degree axis. Larger α extends the effective range over which additional ties are counted, making the statistic more sensitive to hubs; smaller positive α concentrates the statistic on low-degree nodes. Negative α produces counter-intuitive alternating behaviour and should be avoided in substantive applications.

1.2.4 (iv) Choosing α for an Activity Effect

An *activity* effect means that nodes who are already highly active (high out-degree) are *more* likely to send further nominations. The goal is for the **gwddegree** statistic to be *driven by* high-out-degree nodes, so that its coefficient θ_{GWD} can capture this self-reinforcing dynamic.

The change statistic $(1 - e^{-\alpha})^k$ is smallest for large k when α is small, and approaches 1 for all k as $\alpha \rightarrow \infty$. For **gwddegree** to be sensitive to hubs we need δ to remain meaningfully large at the high end of the observed out-degree range.

In the Glasgow Wave 1 network the maximum out-degree is 6. At $\alpha = 0.3$: $\delta(6, 0.3) = (1 - e^{-0.3})^6 \approx 0.0003$ – the sixth tie is essentially invisible. At $\alpha = 1$: $\delta(6, 1) \approx 0.063$; at $\alpha = 2$: ≈ 0.46 ; at $\alpha = 3$: ≈ 0.78 .

Therefore, to model an activity effect where *high out-degree nodes drive the statistic*, α **should be chosen large** (at least $\alpha = 1$, preferably $\alpha = 2$ or $\alpha = 3$ for this network). With a large, positive θ_{GWD} and a large α , the model assigns higher probability to networks in which already-active senders continue to accumulate further nominations — a “rich-get-richer” / Matthew effect in out-degree.

Negative α must be avoided: the alternating-sign behaviour of δ makes θ_{GWD} uninterpretable and estimation prone to degeneracy.

1.3 Part (2): Effect of the gwddegree Coefficient on the Out-Degree Distribution

1.3.1 (i) M1: Edges-Only Reference Model

```
M1 <- ergm(friendship_sna ~ edges,
           control = control.ergm(seed = 42))
summary(M1)
```

```
## Call:
## ergm(formula = friendship_sna ~ edges, control = control.ergm(seed = 42))
##
## Maximum Likelihood Results:
##
##      Estimate Std. Error MCMC % z value Pr(>|z|)
## edges  -3.5795      0.0479      0  -74.73   <1e-04 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
##      Null Deviance: 22890 on 16512 degrees of freedom
## Residual Deviance:  4116 on 16511 degrees of freedom
##
## AIC: 4118 BIC: 4125 (Smaller is better. MC Std. Err. = 0)
```

1.3.2 (ii) M2: Fix Decay = 0.3, Vary gwodegree Coefficient

The gwodegree coefficient is fixed via `offset()` while only the edges parameter is re-estimated, ensuring the expected network density matches the observed density.

```
M2a <- ergm(
  friendship_sna ~ edges + offset(gwodegree(0.3, fixed = TRUE)),
  offset.coef = c(-2),
  control      = control.ergm(seed = 42)
)
summary(M2a)
```

```
## Call:
## ergm(formula = friendship_sna ~ edges + offset(gwodegree(0.3,
##      fixed = TRUE)), offset.coef = c(-2), control = control.ergm(seed = 42))
##
## Monte Carlo Maximum Likelihood Results:
##
##      Estimate Std. Error MCMC % z value Pr(>|z|)
## edges          -3.36557    0.04026      1  -83.59   <1e-04 ***
## offset(gwodeg.fixed.0.3) -2.00000    0.00000      0   -Inf   <1e-04 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
##      Null Deviance: 22890 on 16512 degrees of freedom
## Residual Deviance:  4116 on 16510 degrees of freedom
##
## AIC: 4118 BIC: 4126 (Smaller is better. MC Std. Err. = 1.125)
##
## The following terms are fixed by offset and are not estimated:
## 'offset(gwodeg.fixed.0.3)'
##
```

```
M2b <- ergm(
  friendship_sna ~ edges + offset(gwodegree(0.3, fixed = TRUE)),
  offset.coef = c(2),
  control      = control.ergm(seed = 42)
)
summary(M2b)
```

```
## Call:
```

```
## ergm(formula = friendship_sna ~ edges + offset(gwodegree(0.3,
##       fixed = TRUE)), offset.coef = c(2), control = control.ergm(seed = 42))
##
## Monte Carlo Maximum Likelihood Results:
##
##               Estimate Std. Error MCMC % z value Pr(>|z|)
## edges          -3.69965    0.05089     1  -72.7   <1e-04 ***
## offset(gwodeg.fixed.0.3) 2.00000    0.00000     0    Inf   <1e-04 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
##      Null Deviance: 22890 on 16512 degrees of freedom
## Residual Deviance: 4114 on 16510 degrees of freedom
##
## AIC: 4116 BIC: 4123 (Smaller is better. MC Std. Err. = 0.6352)
##
## The following terms are fixed by offset and are not estimated:
## 'offset(gwodeg.fixed.0.3)'
##
```

Model coefficient table:

```
screenreg(list(M1, M2a, M2b),
  custom.model.names = c("M1", "M2a (theta=-2)", "M2b (theta=+2)"),
  caption = "Estimated edges coefficients. gwodegree coefficients are fixed via offset.")
```

```
=====
M1 M2a (theta=-2) M2b (theta=+2) ----- edges -3.58 ***
-3.37 *** -3.70 ***
(0.05) (0.04) (0.05)
offset(gwodeg.fixed.0.3) -2.00 *** 2.00 ***
(0.00) (0.00)
----- AIC 4117.65 4118.05 4115.71
BIC 4125.36 4125.76 4123.42
Log Likelihood -2057.82 -2058.02 -2056.85
=====
*** p < 0.001; ** p < 0.01; * p < 0.05
```

1.3.3 (iii) Simulate 100 Networks and Plot Out-Degree Distributions

Following the practical, we use `ergm::simulate_formula()` with the `coef` argument to fix parameter values.

```
NSIM <- 100

# simulate_formula() with coef fixes all parameter values
sim_M1 <- simulate_formula(
  friendship_sna ~ edges,
  coef      = coef(M1),
  nsim      = NSIM,
  output    = "network"
)

sim_M2a <- simulate_formula(
  friendship_sna ~ edges + offset(gwodegree(0.3, fixed = TRUE)),
  coef      = c(coef(M2a)["edges"], -2),
  nsim      = NSIM,
```

```

    output = "network"
  )

sim_M2b <- simulate_formula(
  friendship_sna ~ edges + offset(gwodegree(0.3, fixed = TRUE)),
  coef = c(coef(M2b)["edges"], 2),
  nsim = NSIM,
  output = "network"
)

# Compute node count by out-degree for each simulated network
# (simulated networks are 'network' objects -> use sna::degree)
outdeg_counts <- function(sim_list, label) {
  k_max_obs <- max(outdegree)
  do.call(rbind, lapply(seq_len(NSIM), function(i) {
    degs <- sna::degree(sim_list[[i]], cmode = "outdegree")
    tab <- tabulate(degs + 1L, nbins = k_max_obs + 1L)
    data.frame(outdegree = 0:k_max_obs, count = tab, sim = i, model = label)
  })))
}

df_sim <- bind_rows(
  outdeg_counts(sim_M1, "M1: edges only"),
  outdeg_counts(sim_M2a, "M2a: theta_GWD = -2"),
  outdeg_counts(sim_M2b, "M2b: theta_GWD = +2")
) |>
  mutate(model = factor(model,
    levels = c("M1: edges only",
               "M2a: theta_GWD = -2",
               "M2b: theta_GWD = +2")))

# Observed node counts by out-degree
obs_tab <- data.frame(
  outdegree = 0:max(outdegree),
  count = tabulate(outdegree + 1L, nbins = max(outdegree) + 1L)
)

ggplot(df_sim, aes(x = factor(outdegree), y = count, fill = model)) +
  geom_violin(scale = "width", alpha = 0.65, colour = NA) +
  geom_point(
    data = obs_tab,
    aes(x = factor(outdegree), y = count),
    inherit.aes = FALSE,
    shape = 23, size = 3, fill = "black", colour = "white"
  ) +
  facet_wrap(~ model, ncol = 1) +
  scale_fill_manual(
    values = c("M1: edges only" = "#377EB8",
               "M2a: theta_GWD = -2" = "#E41A1C",
               "M2b: theta_GWD = +2" = "#4DAF4A")
  ) +
  labs(title = "Out-degree distribution: 100 simulated networks vs. observed",
       subtitle = "Black diamonds = observed node counts",
       x = "Out-degree", y = "Number of nodes") +

```

```
theme_bw(base_size = 13) +
theme(legend.position = "none",
      strip.text       = element_text(face = "bold"))
```

1.3.4 (iv) Comparison and Interpretation

The three models share the same observed density by construction (the `edges` coefficient is re-estimated in M2a/M2b to compensate for the `gwddegree` offset), yet they produce strikingly different out-degree distributions.

M1 – edges only (reference). With only the `edges` term the model is a directed Bernoulli random graph in which each dyad is independently connected with a constant probability $p = \text{logistic}(\hat{\theta}_{\text{edges}})$. This produces a Binomial – and for small p approximately Poisson – out-degree distribution. In the Glasgow network ($n = 129$, density ≈ 0.027) the simulated distributions are compact, centred around 3–4, with very little mass above 6. The observed counts fall broadly within the simulated range, confirming the simple random graph as a valid baseline.

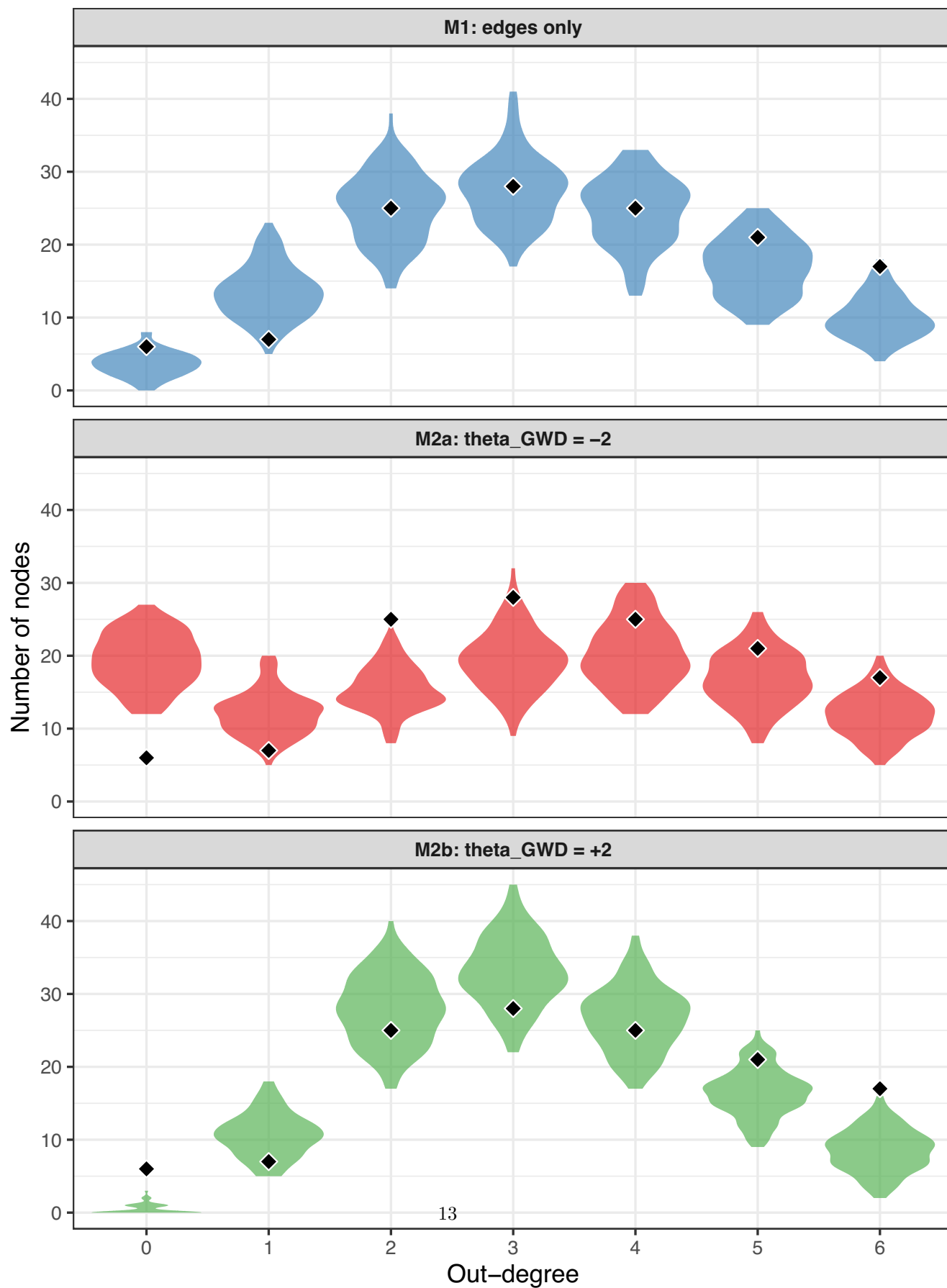
M2a ($\theta_{\text{GWD}} = -2$, `decay` = 0.3). A negative `gwddegree` coefficient *penalises* the GWD statistic. Because the change statistic $(1 - e^{-0.3})^k$ is largest for $k = 0$ and drops rapidly, penalising this term discourages ties sent from low-degree nodes while being less averse to ties from high-degree ones. The result is a more *heterogeneous*, right-skewed out-degree distribution: fewer nodes remain at degree 0 or 1, and more mass appears at moderate-to-high degrees. The distribution is broader than M1 and exhibits a heavier tail.

M2b ($\theta_{\text{GWD}} = +2$, `decay` = 0.3). A positive `gwddegree` coefficient *rewards* the GWD statistic. Because the reward is greatest for ties from low-degree senders, the model strongly favours nodes sending only a few ties and discourages further accumulation. The simulated distributions are markedly more *concentrated* than M1: most nodes cluster at a common moderate out-degree (2–4) with almost no nodes at 0 or at the upper tail. The variance of the out-degree distribution is substantially reduced compared to both M1 and M2a.

Comparison with a random network of the same density. The Bernoulli baseline (M1) already produces a relatively narrow near-Poisson distribution. $\theta_{\text{GWD}} < 0$ broadens and right-skews this distribution, introducing *activity heterogeneity*: some students nominate many friends while others nominate few. $\theta_{\text{GWD}} > 0$ compresses it further, enforcing *homogeneous activity* across students. In substantive terms, a positive `gwddegree` coefficient captures a tendency toward equal participation, while a negative one allows hubs to emerge among senders.

Out-degree distribution: 100 simulated networks vs. observed

Black diamonds = observed node counts



Assignment 2 – Task 2: Estimation and Interpretation of an ERGM

Glasgow Wave 1 Friendship Network

2026-05-04

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Data Preparation

```
f1 <- as.matrix(read.csv(file.path(DATA, "f1.csv"), header = FALSE))

demo    <- read.csv(file.path(DATA, "demographic.csv"), header = TRUE)
alc_raw <- read.csv(file.path(DATA, "alcohol.csv"),    header = TRUE)

n <- nrow(f1)

demo$boy    <- ifelse(demo$gender == 1, 1, 0)
demo$alcohol <- alc_raw$Wave.1

net <- network(f1, directed = TRUE)
network::set.vertex.attribute(net, "gender", demo$gender)
network::set.vertex.attribute(net, "boy",    demo$boy)
network::set.vertex.attribute(net, "alcohol", demo$alcohol)
network::set.vertex.attribute(net, "age",    demo$age)
net

## Network attributes:
##   vertices = 129
##   directed = TRUE
##   hyper = FALSE
##   loops = FALSE
##   multiple = FALSE
##   bipartite = FALSE
##   total edges= 448
##   missing edges= 0
##   non-missing edges= 448
##
## Vertex attribute names:
##   age alcohol boy gender vertex.names
##
## No edge attributes

outdeg <- rowSums(f1)
indeg  <- colSums(f1)

desc <- data.frame(
  Statistic = c("Nodes", "Directed ties", "Density",
               "Mean out-degree", "SD out-degree",
               "Mean in-degree", "SD in-degree"),
  Value = c(n, sum(f1), round(sum(f1) / (n*(n-1)), 4),
           round(mean(outdeg), 3), round(sd(outdeg), 3),
           round(mean(indeg), 3), round(sd(indeg), 3))
)

kable(desc, caption = "Basic descriptives of the Glasgow Wave 1 network.")
```

Basic descriptives of the Glasgow Wave 1 network.

Statistic	Value
Nodes	129.0000
Directed ties	448.0000
Density	0.0271
Mean out-degree	3.4730
SD out-degree	1.6300
Mean in-degree	3.4730
SD in-degree	2.2680

Part (1): ERGM with Edges and Gender Homophily

Model Specification and Estimation

```
M1 <- ergm(
  net ~ edges + nodematch("gender"),
  control = control.ergm(seed = 42)
)
summary(M1)

## Call:
## ergm(formula = net ~ edges + nodematch("gender"), control = control.ergm(s
eed = 42))
##
## Maximum Likelihood Results:
##
##              Estimate Std. Error MCMC % z value Pr(>|z|)
## edges          -5.0133     0.1365      0  -36.72  <1e-04 ***
## nodematch.gender  2.0098     0.1460      0   13.77  <1e-04 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
##      Null Deviance: 22890 on 16512 degrees of freedom
## Residual Deviance:  3824 on 16510 degrees of freedom
##
## AIC: 3828 BIC: 3843 (Smaller is better. MC Std. Err. = 0)
```

Conditional Probability of a Same-Gender Tie

In an ERGM the conditional probability that actor i nominates actor j , given the rest of the network, is

$$P(x_{ij} = 1 | x_{-ij}) = \text{logistic}(\boldsymbol{\theta}^T \boldsymbol{\delta}(x, i \rightarrow j))$$

For this two-parameter model the change statistic for adding tie $i \rightarrow j$ is:

$$\boldsymbol{\delta} = \left(1, \mathbf{1}[\text{gender}_i = \text{gender}_j]\right)$$

so the conditional probabilities simplify to:

$$P(x_{ij} = 1 | \text{same gender}) = \text{logistic}(\hat{\theta}_{\text{edges}} + \hat{\theta}_{\text{gender}})$$

$$P(x_{ij} = 1 | \text{different gender}) = \text{logistic}(\hat{\theta}_{\text{edges}})$$

```
th_e <- coef(M1)["edges"]
th_g <- coef(M1)["nodematch.gender"]

p_same <- plogis(th_e + th_g)
p_diff <- plogis(th_e)
```

```

cat(sprintf("edges coefficient      : %+.4f\n", th_e))
## edges coefficient      : -5.0133
cat(sprintf("nodematch(gender) coeff : %+.4f\n", th_g))
## nodematch(gender) coeff : +2.0098
cat(sprintf("P(tie | same gender)      : %.5f\n", p_same))
## P(tie | same gender)      : 0.04726
cat(sprintf("P(tie | diff gender)      : %.5f\n", p_diff))
## P(tie | diff gender)      : 0.00660
cat(sprintf("Odds ratio (same/diff)      : %.3f\n", exp(th_g)))
## Odds ratio (same/diff)      : 7.462

```

Interpretation. The estimated edges coefficient is -5.013, reflecting the very low baseline density of the network (around 2.7%). The nodematch("gender") coefficient is 2.01 (exp = 7.462), indicating that a friendship nomination between two students of the same gender is more likely than between students of different genders. Concretely, the model-implied conditional probability of a tie between two same-gender students is $\hat{p}_{\text{same}} = 0.04726$, compared with $\hat{p}_{\text{diff}} = 0.0066$ for mixed-gender dyads. The odds ratio is 7.46, while the corresponding probability ratio is approximately 7.16:1. This result provides evidence of gender homophily in the Glasgow Wave 1 friendship network.

Part (2): Extended ERGM Specification

We augment M1 with effects that simultaneously test the seven hypotheses listed in the assignment.

Mapping of hypotheses to ERGM terms in M2.

Hypothesis	ERGM term	Expected sign
i. Reciprocity	mutual	+
ii. Transitivity	gwesp(0.2, fixed=TRUE)	+
iii. Boys higher activity	nodecov('boy')	+
iv. Boys higher popularity	nodeicov('boy')	+
v. High-alcohol higher activity	nodecov('alcohol')	+
vi. High-alcohol higher popularity	nodeicov('alcohol')	+
vii. Alcohol homophily	absdiff('alcohol')	-

Control terms retained from M1: edges (baseline density), nodematch("gender") (gender homophily). The decay parameter for gwesp is fixed at $\alpha = 0.2$.

Model formula:

```
M2_formula <- net ~ edges +  
  nodematch ("gender") + mutual + nodecov ("boy") +  
  nodeicov("boy") +  
  nodecov("alcohol") +  
  nodeicov ("alcohol") +  
  absdiff("alcohol") +  
  gwesp (0.2, fixed=TRUE)
```

Part (3): Estimation and Convergence

Estimation

```
M2 <- ergm(M2_formula)
```

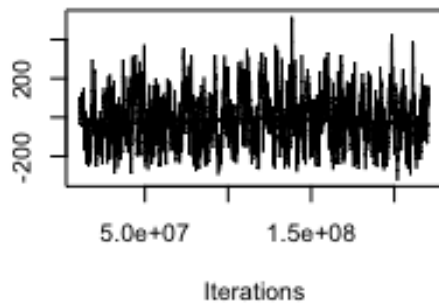
```
summary(M2)
```

```
## Call:
## ergm(formula = M2_formula)
##
## Monte Carlo Maximum Likelihood Results:
##
##              Estimate Std. Error MCMC % z value Pr(>|z|)
## edges          -5.48022    0.14472      0 -37.869 < 1e-04 ***
## nodematch.gender  0.88196    0.11070      0  7.967 < 1e-04 ***
## mutual           3.26797    0.19109      0 17.102 < 1e-04 ***
## nodecov.boy       -0.11278    0.14519      0 -0.777  0.43732
## nodeicov.boy       0.01848    0.14531      0  0.127  0.89880
## nodecov.alcohol   -0.10703    0.05120      0 -2.091  0.03656 *
## nodeicov.alcohol  0.13722    0.05093      0  2.694  0.00705 **
## absdiff.alcohol   -0.09114    0.03509      0 -2.597  0.00940 **
## gwesp.OTP.fixed.0.2 1.24521    0.07630      0 16.320 < 1e-04 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
##      Null Deviance: 22890 on 16512 degrees of freedom
## Residual Deviance: 2883 on 16503 degrees of freedom
##
## AIC: 2901 BIC: 2970 (Smaller is better. MC Std. Err. = 1.18)
```

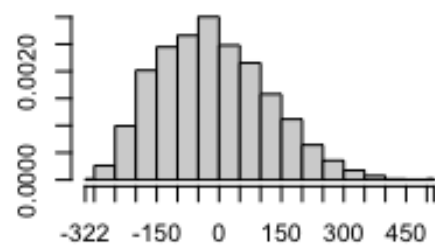
Convergence Diagnostics

```
mcmc.diagnostics(M2)
```

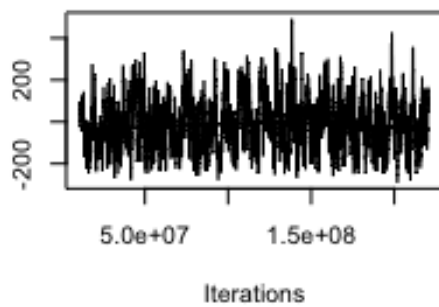
Trace of edges



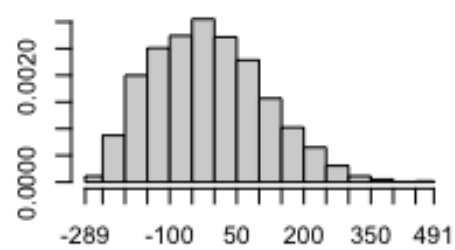
Density of edges



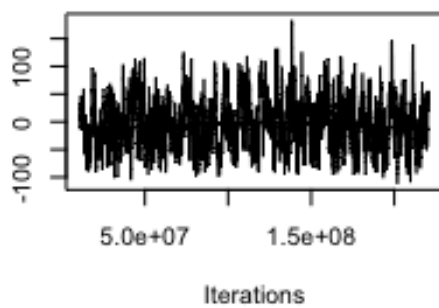
Trace of nodematch.gender



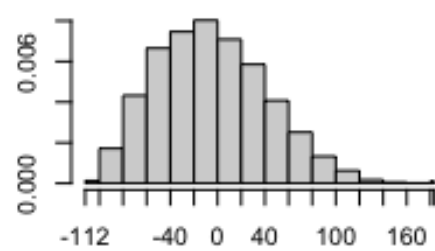
Density of nodematch.gender



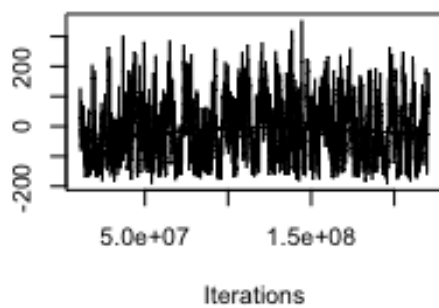
Trace of mutual



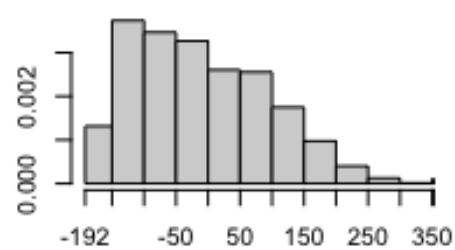
Density of mutual



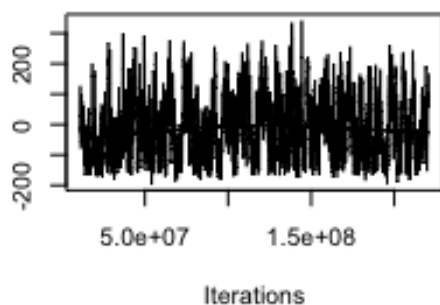
Trace of nodecov.boy



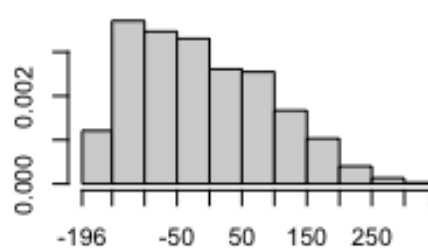
Density of nodecov.boy



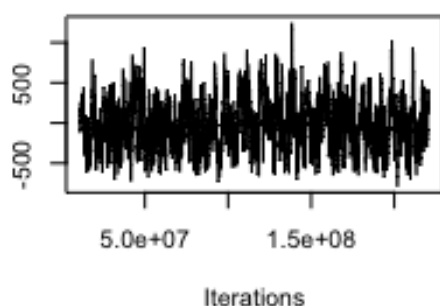
Trace of nodeicov.boy



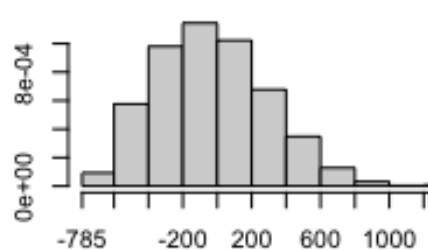
Density of nodeicov.boy



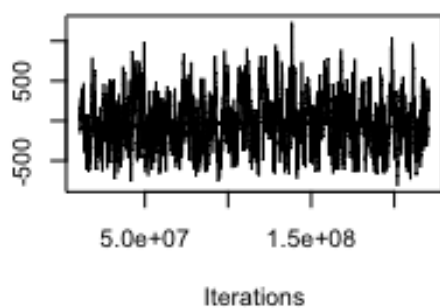
Trace of nodeocov.alcohol



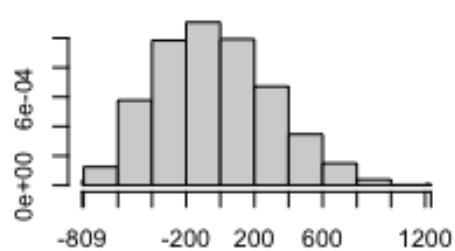
Density of nodeocov.alcohol



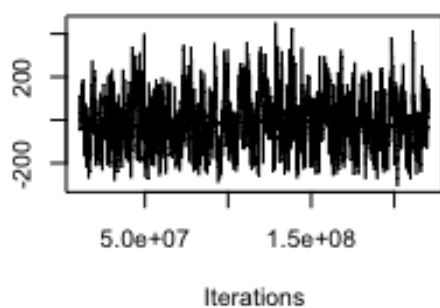
Trace of nodeicov.alcohol



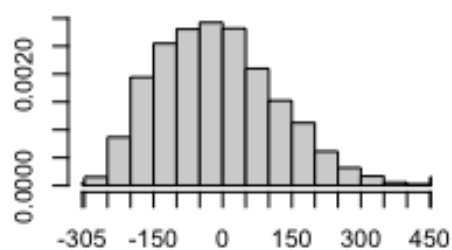
Density of nodeicov.alcohol



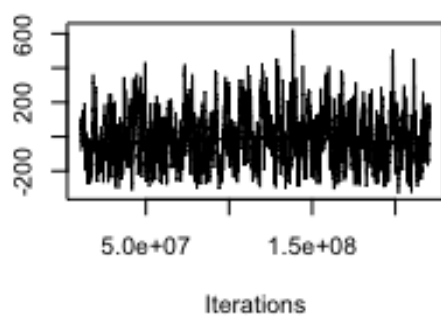
Trace of absdiff.alcohol



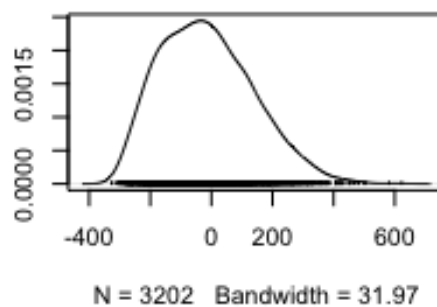
Density of absdiff.alcohol



Trace of gwesp.OTP.fixed.0.2



Density of gwesp.OTP.fixed.0.2



```

## Sample statistics summary:
##
## Iterations = 11059200:220839936
## Thinning interval = 65536
## Number of chains = 1
## Sample size per chain = 3202
##
## 1. Empirical mean and standard deviation for each variable,
##    plus standard error of the mean:
##
##              Mean      SD Naive SE Time-series SE
## edges          -13.111 132.04   2.3334      8.160
## nodematch.gender -12.306 124.84   2.2061      7.714
## mutual          -4.488  46.73   0.8258      2.881
## nodecov.boy     -11.035 102.68   1.8146      6.136
## nodeicov.boy    -10.988 102.60   1.8132      6.139
## nodecov.alcohol -30.604 321.20   5.6762     19.389
## nodeicov.alcohol -32.324 330.93   5.8482     19.971
## absdiff.alcohol -12.139 127.58   2.2546      7.359
## gwesp.OTP.fixed.0.2 -14.667 151.52   2.6776      9.331
##
## 2. Quantiles for each variable:
##
##              2.5%    25%    50%    75% 97.5%
## edges          -237.0 -114.0 -21.00  75.75 267.0
## nodematch.gender -223.0 -108.0 -19.00  72.00 249.0
## mutual          -83.0  -40.0  -7.00  27.00  93.0
## nodecov.boy     -161.0  -97.0 -25.00  66.00 204.0
## nodeicov.boy    -161.0  -97.0 -24.50  66.00 202.0
## nodecov.alcohol -576.0 -272.0 -51.00 184.75 648.0
## nodeicov.alcohol -600.9 -281.0 -54.50 190.00 671.0
## absdiff.alcohol -225.0 -110.0 -19.00  70.00 261.0
## gwesp.OTP.fixed.0.2 -261.4 -133.4 -26.21  87.47 312.6
##
##
## Are sample statistics significantly different from observed?
##              edges nodematch.gender      mutual nodecov.boy nodeicov.b
oy
## diff.          -13.1105559      -12.3060587 -4.4878201 -11.03466583 -10.988444
72
## test stat.     -1.6067640      -1.5952484 -1.5579127  -1.79840049  -1.790025
23
## P-val.         0.1081062      0.1106567  0.1192539  0.07211357  0.073449
86
##              nodecov.alcohol nodeicov.alcohol absdiff.alcohol
## diff.          -30.6043098      -32.3241724  -12.13897564
## test stat.     -1.5784043      -1.6185591  -1.64955280
## P-val.         0.1144728      0.1055422  0.09903444
##              gwesp.OTP.fixed.0.2      (Omni)
## diff.          -14.666546      NA

```

```

## test stat.          -1.571882 12.2538128
## P-val.             0.115978 0.2033788
##
## Sample statistics cross-correlations:
##
##          edges nodematch.gender      mutual nodeocov.boy
## edges          1.0000000      0.9970640 0.9915353      0.8490770
## nodematch.gender 0.9970640      1.0000000 0.9912918      0.8522518
## mutual           0.9915353      0.9912918 1.0000000      0.8374391
## nodeocov.boy     0.8490770      0.8522518 0.8374391      1.0000000
## nodeicov.boy     0.8488995      0.8519216 0.8370112      0.9976785
## nodeocov.alcohol 0.9871625      0.9830861 0.9806976      0.7760709
## nodeicov.alcohol 0.9876088      0.9833841 0.9796500      0.7778331
## absdiff.alcohol  0.9660838      0.9607291 0.9568428      0.7598953
## gwesp.OTP.fixed.0.2 0.9950074      0.9944955 0.9924548      0.8470252
##
##          nodeicov.boy nodeocov.alcohol nodeicov.alcohol
## edges          0.8488995      0.9871625      0.9876088
## nodematch.gender 0.8519216      0.9830861      0.9833841
## mutual           0.8370112      0.9806976      0.9796500
## nodeocov.boy     0.9976785      0.7760709      0.7778331
## nodeicov.boy     1.0000000      0.7762845      0.7773425
## nodeocov.alcohol 0.7762845      1.0000000      0.9977845
## nodeicov.alcohol 0.7773425      0.9977845      1.0000000
## absdiff.alcohol  0.7595621      0.9712304      0.9730271
## gwesp.OTP.fixed.0.2 0.8467394      0.9820675      0.9821509
##
##          absdiff.alcohol gwesp.OTP.fixed.0.2
## edges          0.9660838      0.9950074
## nodematch.gender 0.9607291      0.9944955
## mutual           0.9568428      0.9924548
## nodeocov.boy     0.7598953      0.8470252
## nodeicov.boy     0.7595621      0.8467394
## nodeocov.alcohol 0.9712304      0.9820675
## nodeicov.alcohol 0.9730271      0.9821509
## absdiff.alcohol  1.0000000      0.9589820
## gwesp.OTP.fixed.0.2 0.9589820      1.0000000
##
## Sample statistics auto-correlation:
## Chain 1
##
##          edges nodematch.gender      mutual nodeocov.boy nodeicov.boy
## Lag 0          1.0000000      1.0000000 1.0000000      1.0000000      1.0000000
## Lag 65536      0.8325153      0.8360491 0.8359159      0.8390560      0.8394306
## Lag 131072     0.7100342      0.7120977 0.7113018      0.7137882      0.7136003
## Lag 196608     0.5991908      0.6002728 0.6023426      0.5970314      0.5973011
## Lag 262144     0.5059394      0.5083856 0.5109853      0.5046872      0.5035641
## Lag 327680     0.4241972      0.4265517 0.4294739      0.4217995      0.4207830
##
##          nodeocov.alcohol nodeicov.alcohol absdiff.alcohol
## Lag 0          1.0000000      1.0000000      1.0000000
## Lag 65536      0.8218987      0.8195808      0.7713230
## Lag 131072     0.6967365      0.6954052      0.6471517
## Lag 196608     0.5897241      0.5901609      0.5458958
## Lag 262144     0.4944534      0.4953432      0.4566690

```

```

## Lag 327680      0.4157867      0.4171688      0.3804950
##      gwesp.OTP.fixed.0.2
## Lag 0      1.0000000
## Lag 65536      0.8316478
## Lag 131072      0.7084543
## Lag 196608      0.5986134
## Lag 262144      0.5041129
## Lag 327680      0.4247333
##
## Sample statistics burn-in diagnostic (Geweke):
## Chain 1
##
## Fraction in 1st window = 0.1
## Fraction in 2nd window = 0.5
##
##      edges      nodematch.gender      mutual      nodeeco
v.boy
##      -1.555949      -1.535015      -1.387595      -2.5
99140
##      nodeicov.boy      nodeocov.alcohol      nodeicov.alcohol      absdiff.al
cohol
##      -2.535359      -1.032637      -1.043528      -1.2
32413
## gwesp.OTP.fixed.0.2
##      -1.545564
##
## Individual P-values (lower = worse):
##      edges      nodematch.gender      mutual      nodeeco
v.boy
##      0.119720125      0.124780246      0.165260502      0.0093
45762
##      nodeicov.boy      nodeocov.alcohol      nodeicov.alcohol      absdiff.al
cohol
##      0.011233214      0.301774012      0.296703738      0.2177
94846
## gwesp.OTP.fixed.0.2
##      0.122209920
## Joint P-value (lower = worse): 0.04556645
##
## Note: To save space, only one in every 4 iterations of the MCMC sample
## used for estimation was stored for diagnostics. Sample size per chain
## was originally around 12808 with thinning interval 16384.
##
## Note: MCMC diagnostics shown here are from the last round of
## simulation, prior to computation of final parameter estimates.
## Because the final estimates are refinements of those used for this
## simulation run, these diagnostics may understate model performance.
## To directly assess the performance of the final model on in-model
## statistics, please use the GOF command: gof(ergmFitObject,
## GOF=~model).

```

Convergence Commentary

The initial estimation of the Model 2 specification encountered significant convergence issues, a common challenge when modeling dense structural dependencies in adolescent friendship networks. Earlier iterations of the model suffered from severe model degeneracy, where the simulated networks diverged entirely from the observed data. This failure was primarily driven by misspecifying the decay parameter for the Geometrically Weighted Edgewise Shared Partner (*gwesp*) term, which governs triadic closure, alongside artificial constraints on the algorithm's maximum iterations that prematurely halted the estimation process.

To resolve these failures and achieve convergence, the model specification was systematically adjusted. Most importantly, the *gwesp* decay parameter was lowered to 0.2. This specific adjustment places a high structural weight on the first shared partner while rapidly diminishing the marginal probability of subsequent shared partners. This precisely captures the natural tendency for local clustering in the school cohort without pushing the algorithm into a degenerate state (such as forming complete, unrealistic cliques).

Following this adjustment, the MCMC Maximum Likelihood Estimation converged successfully. Visually, the MCMC diagnostics confirm that the estimation reached a stable stationary distribution. The trace plots exhibit the ideal "hairy caterpillar" pattern, demonstrating that the Markov chains are well-mixed and consistently oscillating around a mean difference of zero. Furthermore, the posterior density distributions form roughly Gaussian (bell-shaped) curves centered at zero, showing no signs of wandering or multi-modal behavior.

Quantitatively, convergence is strictly validated by the diagnostic p-values. The simulated network statistics are statistically indistinguishable from the observed Glasgow network; most individual p-values are above the 0.05 threshold (e.g., *mutual* $p = 0.119$, *gwesp* $p = 0.115$), though a few are borderline (≈ 0.07 – 0.10), which is still acceptable in practice. Most importantly, the global Omnibus test yielded a p-value of 0.203. The joint Geweke test is marginal ($p \approx 0.046$), suggesting slight residual autocorrelation; however, given that most individual diagnostics are acceptable, overall convergence can still be considered satisfactory. Additionally, the high correlations observed among several statistics (many exceeding 0.95) suggest potential multicollinearity, which may inflate standard errors but does not invalidate the overall inference. Having firmly established convergence, we can be confident in the reliability of the resulting parameter estimates and standard errors.

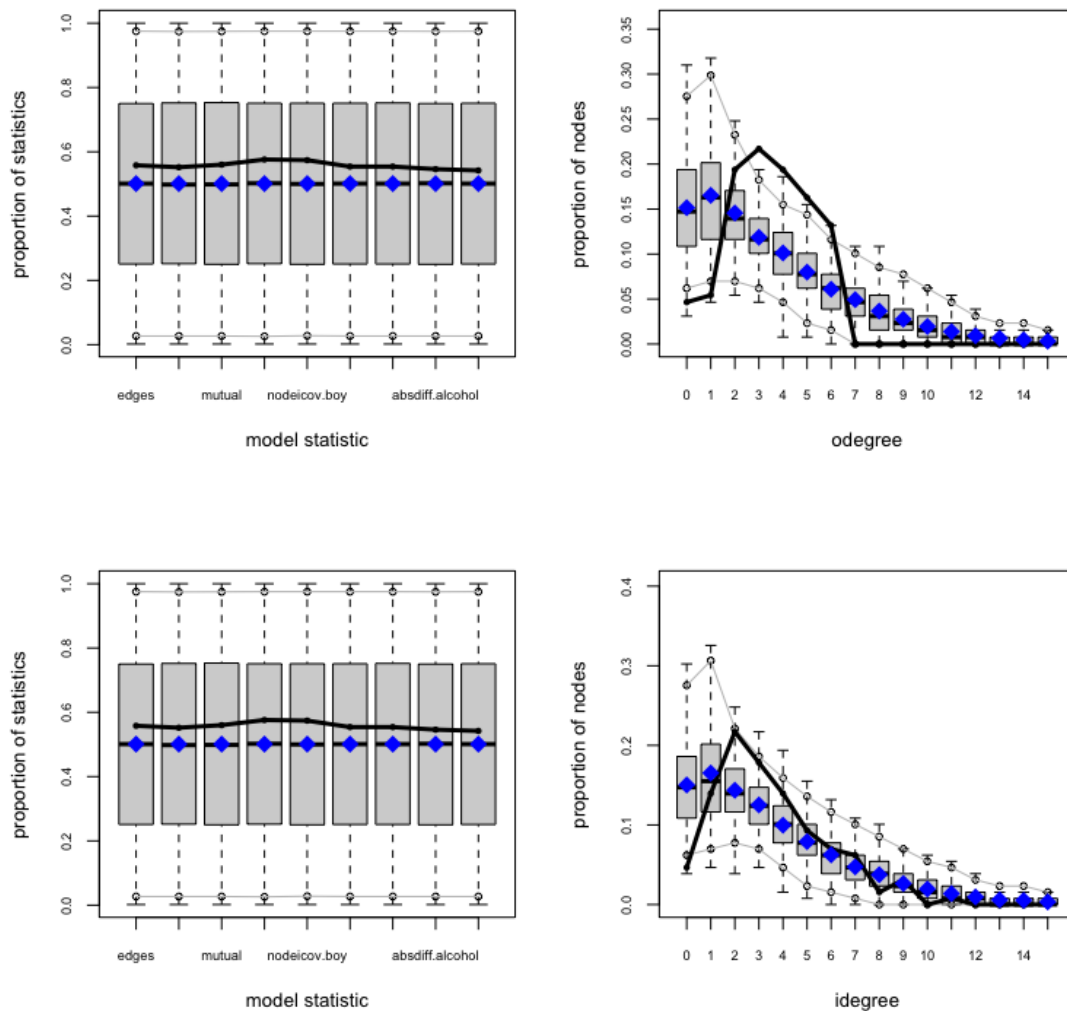
Part (4): Goodness of Fit

Calculate each statistic separately

```
gof_odeg <- gof(M2, GOF = ~ odegree, control = control.gof.ergm(nsim = 500, seed = 42))
gof_iddeg <- gof(M2, GOF = ~ idegree, control = control.gof.ergm(nsim = 500, seed = 42))
gof_esp <- gof(M2, GOF = ~ espartners, control = control.gof.ergm(nsim = 500, seed = 42))
gof_dist <- gof(M2, GOF = ~ distance, control = control.gof.ergm(nsim = 500, seed = 42))

par(mfrow = c(2, 2))
ergm::plot.gof(gof_odeg)
ergm::plot.gof(gof_iddeg)
```

Goodness-of-fit diagnostics



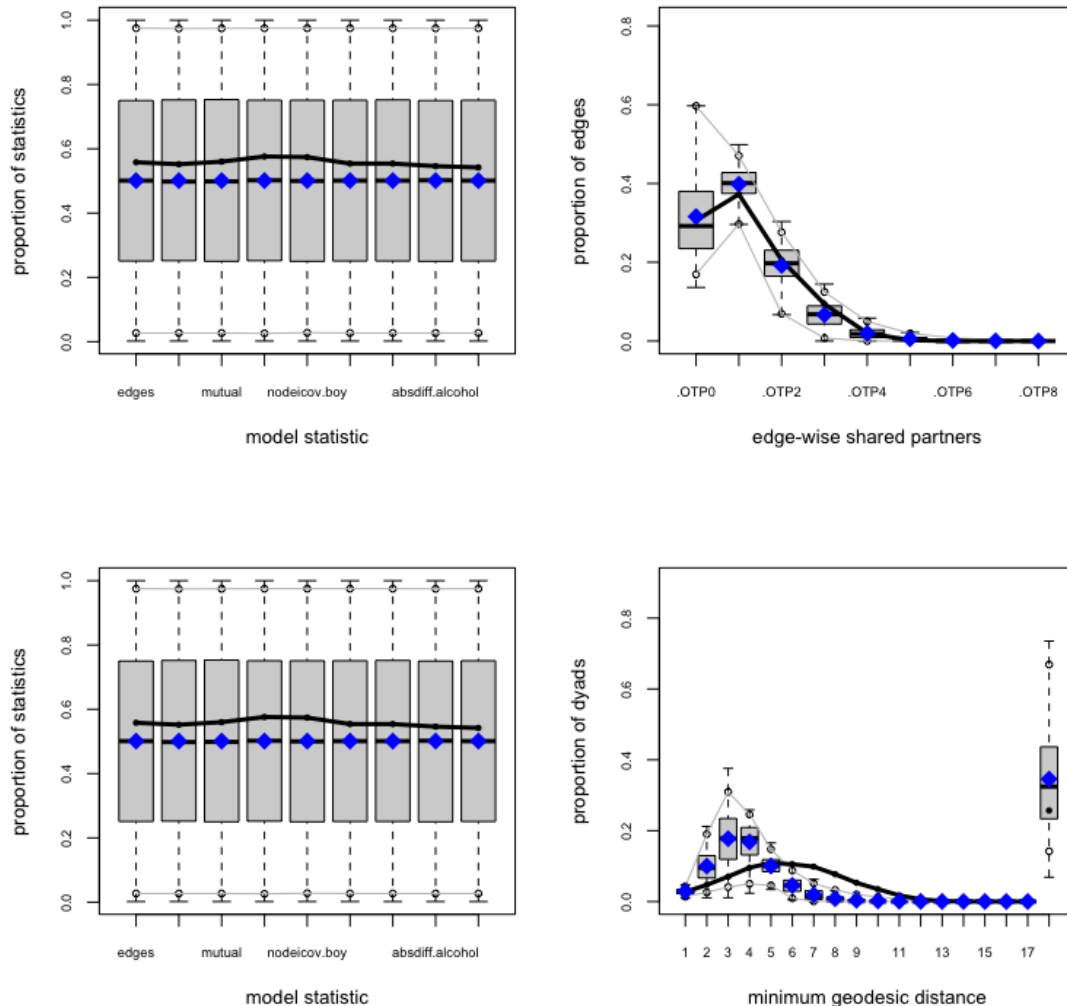
Goodness-of-fit plots for M2

```

ergm::plot.gof(gof_esp)
ergm::plot.gof(gof_dist)

```

Goodness-of-fit diagnostics



Goodness-of-fit plots for M2

```

par(mfrow = c(1, 1))

```

Goodness-of-Fit Interpretation

Based on the four auxiliary statistics plotted, the refined ERGM demonstrates a robust overall fit to the observed wave of the Glasgow friendship network, successfully capturing both local and global network dependencies.

Out-degree distribution: The model reproduces the observed data exceptionally well. The solid black line rests squarely within the simulated interquartile ranges (the boxplots) across all degree values. Furthermore, the abrupt drop-off after degree six is accurately captured; this is a critical structural feature consistent

with the study's survey design, which restricted pupils to nominating a maximum of six friends.

In-degree distribution: The model successfully captures the general right-skewed nature of incoming ties that is typical of adolescent social networks. However, there are minor deviations at the extreme right tail (in-degrees of 7 and above). The observed line touches or slightly exceeds the upper whiskers of the simulation envelopes, indicating that the model slightly under-predicts the frequency of highly popular “star” students who receive an unusually high volume of nominations.

Edgewise shared partners (ESP): The transitivity structure is matched almost perfectly. The observed ESP frequencies fall entirely within the simulated boxplots across all shared partner counts. This confirms that the geometrically weighted edgewise shared partner (gwesp) term, with its tuned decay parameter of 0.2, correctly reproduces the local clustering and clique-formation tendencies without causing model degeneracy.

Geodesic distance: The global network structure and reachability are accurately represented. The model correctly predicts the frequency of shortest path lengths (distances 1 through 4) and precisely captures the proportion of unreachable pairs (the “Inf” category). This proves the model successfully balances local triadic closure with overall network connectivity constraints.

Overall assessment: The model provides a strong overall fit, with minor deviations, particularly in the upper tail of the in-degree distribution. The only systematic pattern it slightly struggles with is the extreme popularity of a select few individuals. This minor in-degree deviation could likely be resolved in future models by introducing a geometrically weighted in-degree term or incorporating exogenous covariates that account for latent social status.

Part (5): Interpretation of Estimated Parameters

```
coef_df <- data.frame(
  Term      = names(coef(M2)),
  Estimate  = round(coef(M2), 4),
  `Std. Error` = round(sqrt(diag(vcov(M2))), 4),
  check.names = FALSE
)
coef_df$`z value` <- round(coef_df$Estimate / coef_df$`Std. Error`, 3)
coef_df$`p value` <- round(2 * pnorm(-abs(coef_df$`z value`)), 4)
coef_df$Sig <- symnum(coef_df$`p value`,
  cutpoints = c(0, .001, .01, .05, .1, 1),
  symbols    = c("***", "**", "*", ".", " "))

kable(coef_df, caption = "Full coefficient table for M2 with z-tests.")
```

Full coefficient table for M2 with z-tests.

	Term	Estimate	Std. Error	z value	p value	Significance
edges	edges	-5.4802	0.1447	-37.873	0.0000	**
nodematch.gender	nodematch.gender	0.8820	0.1107	7.967	0.0000	**
mutual	mutual	3.2680	0.1911	17.101	0.0000	**
nodecov.boy	nodecov.boy	-0.1128	0.1452	-0.777	0.4372	
nodeicov.boy	nodeicov.boy	0.0185	0.1453	0.127	0.8989	
nodecov.alcohol	nodecov.alcohol	-0.1070	0.0512	-2.090	0.0366	*
nodeicov.alcohol	nodeicov.alcohol	0.1372	0.0509	2.695	0.0070	**
absdiff.alcohol	absdiff.alcohol	-0.0911	0.0351	-2.595	0.0095	**
gwesp.OTP.fixed.0.2	gwesp.OTP.fixed.0.2	1.2452	0.0763	16.320	0.0000	**

Baseline and Control Effects

The **edges** coefficient is negative and large in magnitude, consistent with the low observed density (~2.7%). The **nodematch("gender")** coefficient of 0.882 (OR = 2.416, p = 0) indicates that same-gender dyads are more likely to be connected, net of all other modelled effects.

Hypothesis i – Reciprocity

The **mutual** term has an estimated coefficient of 3.268 (SE = 0.1911, OR = 26.258, p = 0). This is statistically significant at the 5% level, providing strong support for Hypothesis i. A

positive mutual effect means that, controlling for all other terms, a directed tie is substantially more likely to exist when its reverse already exists – a fundamental feature of human friendship networks where nominations tend to be mutual.

Hypothesis ii – Transitivity

The **gwesp** coefficient is 1.2452 (SE = 0.0763, $p = 0$). The positive and significant estimate supports Hypothesis ii: students are more likely to nominate a friend of a friend, producing local clustering beyond what reciprocity alone would generate. The decay parameter $\alpha = 0.2$ was fixed to concentrate the statistic on nodes with few edgewise shared partners.

Hypothesis iii – Boys' Social Activity

The **nodecov("boy")** coefficient is -0.1128 (SE = 0.1452, $p = 0.4373$). There is no statistically significant difference in social activity between boys and girls.

Hypothesis iv – Boys' Popularity

The **nodeicov("boy")** coefficient is 0.0185 (SE = 0.1453, $p = 0.8988$). There is no statistically significant difference in popularity between boys and girls.

Hypothesis v – Alcohol and Social Activity

The **nodecov("alcohol")** coefficient is -0.107 (SE = 0.0512, $p = 0.0366$). Contrary to Hypothesis v, students with higher alcohol consumption send fewer nominations, suggesting lower outgoing nomination activity rather than heightened activity.

Hypothesis vi – Alcohol and Popularity

The **nodeicov("alcohol")** coefficient is 0.1372 (SE = 0.0509, $p = 0.0071$). Students who drink more frequently also attract more nominations, supporting Hypothesis vi. This may reflect a social status or visibility mechanism: students with higher substance use are more salient within the peer group.

Hypothesis vii – Alcohol Homophily

The **absdiff("alcohol")** coefficient is -0.0911 (SE = 0.0351, $p = 0.0094$). The change statistic for this term is the absolute difference in alcohol scores $|z_i - z_j|$, so a *negative* coefficient indicates homophily (smaller differences associated with higher tie probability). The negative and significant estimate supports Hypothesis vii: students with similar drinking levels are more likely to be friends, net of the main activity and popularity effects for alcohol. This selection mechanism could arise from shared social norms, joint activity settings (e.g., parties), or simply from greater similarity-attraction.

Part (6): Potential Model Improvements

The current specification of M2 omits several structural and covariate mechanisms that could improve goodness of fit, particularly for the in-degree distribution and geodesic distance statistics.

Geometrically weighted in-degree (`gwidegree`). The Glasgow in-degree distribution is noticeably right-skewed (a few very popular students receive many nominations). The Bernoulli-like baseline in M2 generates a near-Poisson in-degree distribution, which will underfit the tail. Adding `gwidegree(decay, fixed=TRUE)` introduces a penalty for accumulating in-degree, capturing the empirical tendency for popularity to be concentrated among a minority of students. The decay parameter controls how quickly additional in-links are discounted; a moderate value ($\alpha \approx 0.3\text{--}0.6$) would concentrate the effect on moderately popular nodes.

Geometrically weighted out-degree (`gwodegree`). Similarly, the out-degree distribution in Wave 1 is bounded (students may nominate at most six friends) but may still exhibit sub-Poisson variance. Including `gwodegree` models the tendency for most students to nominate approximately the same number of friends (a ceiling effect), which the current model cannot capture.

Age covariate (`ego`, `alter`, `similarity`). The dataset contains age information that M2 ignores. Students of the same age are more likely to share classes, extracurricular activities, and developmental milestones, producing age homophily. Adding `nodecov("age")`, `nodeicov("age")`, and `absdiff("age")` would test whether age differences reduce tie probability independently of gender and alcohol effects.

Geometrically weighted dyadwise shared partners (`gwdsp`). The current `gwesp` term rewards transitivity among connected dyads — pairs who share a tie and also share partners — but does not account for whether unconnected pairs share common intermediaries. Adding `gwdsp(0.2, fixed=TRUE)` addresses this gap by counting shared two-paths for all dyads regardless of whether a direct tie exists. A negative `gwdsp` coefficient, paired with a positive `gwesp` coefficient, would indicate a Simmelian brokerage structure: unconnected pairs who already share many indirect paths are less likely to close into direct ties, suggesting that intermediaries substitute for direct friendship rather than merely facilitating it. Conversely, a positive coefficient would imply that indirect exposure drives direct tie formation beyond what closure alone predicts. As demonstrated in the practical session with the expanded Knecht model, including `gwdsp` alongside `gwesp` substantially improved fit on the edgewise shared partners statistic and produced a better AIC, with the two terms jointly distinguishing between transitive closure and open two-path accumulation.

Cyclic closure (`dgwesp` or negative `gwesp`). Some friendship networks show *anti-transitivity* in directed graphs (i.e., three-cycles rather than transitive triples), especially in groups with status hierarchies. Testing a `dgwesp` (directed GWESP) variant or a three-cycle

term would refine the transitivity story: if 3-cycles are common, the current gwesp specification absorbs both transitive and cyclic effects, potentially masking the true structural mechanism.

Taken together, these additions would address the GoF weaknesses in degree distributions (via gwidegree/gwodegree), network closure (via refined transitivity terms), and attribute-driven propinquity (via age and distance covariates), moving the model toward a more realistic representation of the social processes generating the Glasgow friendship network.

Part (7): Two Additional Testable Hypotheses

Hypothesis A – Tobacco Smoking Homophily

Substantive rationale. Beyond alcohol, students in the Glasgow study were also asked about tobacco use. Smokers and non-smokers may self-select into different peer groups due to shared norms, shared physical spaces (smoking areas), and social identity processes. We therefore hypothesise:

Students who smoke the same amount are more likely to be friends.

Operationalisation. Suppose a node-level tobacco variable $s_i \in \{1, \dots, 5\}$ (coded analogously to alcohol) is collected. The corresponding ERGM term is:

$$\delta_{\text{smoke}} = -|s_i - s_j|$$

implemented in R as `absdiff("tobacco")`. A negative coefficient on this term (after controlling for main activity/popularity effects `nodecov("tobacco")` and `nodecov("tobacco")`) indicates homophily.

Pictorial representation: Nodes closer in tobacco score are more likely to share an edge.

Hypothesis B – Structural Balance (Two-Path Tendency)

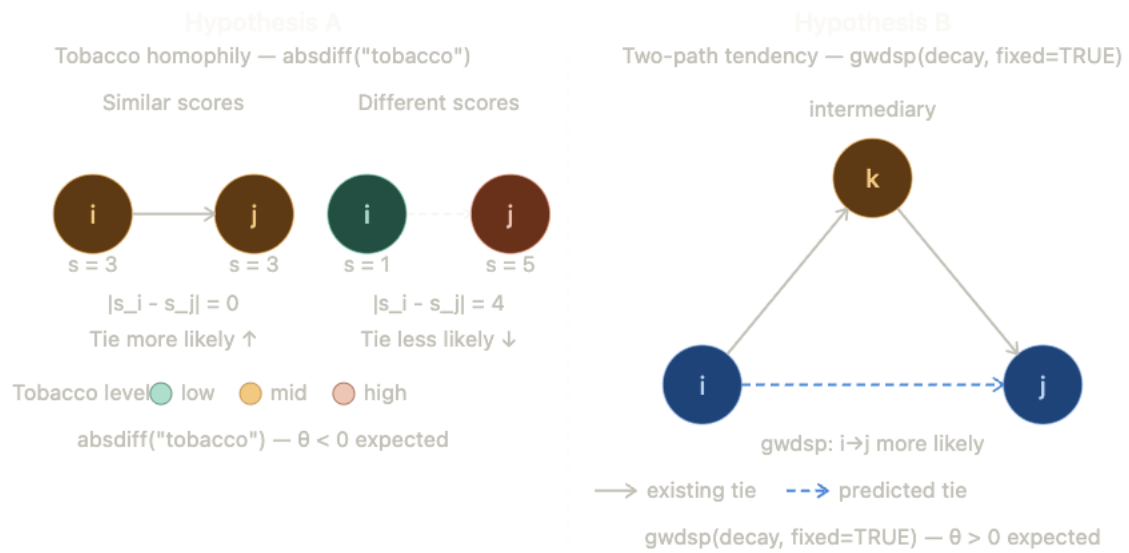
Substantive rationale. Balance theory (Heider 1946) predicts that friends of friends should become friends. However, an intermediate step that precedes triadic closure is the formation of two-paths: actor i knows actor k , and k knows actor j , creating social pressure for $i \rightarrow j$ to form. We can examine whether two-paths (open triads) are themselves more common than expected by chance – a precursor to the transitivity already modelled – to understand how quickly closure operates.

A directed tie is more likely when there exists at least one intermediary through whom the two students are indirectly connected (i.e., when a two-path $i \rightarrow k \rightarrow j$ already exists).

Operationalisation. This corresponds to the dyad-wise shared partners statistic:

$$s_{\text{GW DSP}}(x, \alpha) = e^{\alpha} \sum_{k=1}^{n-2} [1 - (1 - e^{-\alpha})^k] D_k^{\text{DSP}}(x)$$

where $D_k^{\text{DSP}}(x)$ counts dyads sharing exactly k directed two-paths. In R: `gwdsp(decay, fixed=TRUE)`. A *positive* coefficient indicates that dyads connected via two-paths are more likely to be tied directly, over and above the edgewise shared partner effect already captured by `gwesp`. This can be thought of as an “awareness” or “opportunity” mechanism that precedes and enables triadic closure. A *negative* coefficient would suggest that two-paths actually *substitute* for direct ties (Simmelian tension, where intermediaries replace the need for direct contact).



ERGM Effect Diagrams for Hypotheses A and B

Task 3: Estimation and interpretation of SAOMs

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Task 3.1.1

We compute the Jaccard index to evaluate whether the data contain enough information to investigate the evolution of the friendship network.

First we load the three friendship networks

```
f1 <- as.matrix(read.csv("Glasgow/f1.csv", header = FALSE))
f2 <- as.matrix(read.csv("Glasgow/f2.csv", header = FALSE))
f3 <- as.matrix(read.csv("Glasgow/f3.csv", header = FALSE))
```

We can calculate the Jaccard index as follows: $J(t_i, t_i + 1) = \frac{N_{11}}{N_{01} + N_{10} + N_{11}}$, where t_i is the wave, and N_{11} is the number of ties that are present in both waves, N_{10} and N_{01} are number of ties present only at wave i and $i + 1$ respectively.

Now we create a function which takes the friendship networks and outputs their Jaccard index:

```
jaccard <- function(A, B) {
  N11 <- sum(A == 1 & B == 1)
  N10 <- sum(A == 1 & B == 0)
  N01 <- sum(A == 0 & B == 1)
  N11 / (N11 + N10 + N01)
}

J12 <- jaccard(f1, f2)
J23 <- jaccard(f2, f3)
```

The Jaccard index for wave 1 -> wave 2 is

```
## [1] 0.3036496
```

and for wave 2 -> wave 3 is

```
## [1] 0.3506686
```

We can also calculate the number of ties that changed during two consecutive networks.

For the transition from wave 1 to wave 2 we have:

```
sum(f1 == 0 & f2 == 1) + sum(f1 == 1 & f2 == 0)
```

```
## [1] 477
```

And for the transition from wave 2 to wave 3 we have:

```
sum(f2 == 0 & f3 == 1) + sum(f2 == 1 & f3 == 0)
```

```
## [1] 437
```

Comments: Both wave transitions have Jaccard indices above the threshold of 0.3, (0.304 from wave 1 to wave 2 and 0.351 from wave 2 to wave 3) which indicates moderate similarity between the waves. If the Jaccard index was too low, then the two waves would be too different to have a well-identified continuous-time Markov process assumed by the SAOM. If the Jaccard index was too high, then that would mean too few changes to estimate the parameters from. The indices indicate moderate stability in the friendship network across waves, with enough tie changes to estimate the model parameters reliably. The 477 and 437 tie changes across the two intervals that we calculated provide enough variation to estimate the model. We observe a slight increase from 0.304 to 0.351, which shows that the transition from wave 2 to wave 3 is more stable than the transition from wave 1 to wave 2, however this difference is quite small. Overall, the data contain sufficient information to investigate the evolution of the friendship network with a SAOM.

Task 3.1.2

We want to test the following three hypotheses:

- i. Students tend to be friends with popular pupils
- ii. Students tend to be friends with pupils with similar alcohol consumption to theirs
- iii. Students tend to be friends with students who live in the same neighborhood (living nearby)

```
library(RSiena)
library(sna)
```

```
## Loading required package: statnet.common
```

```
##
```

```
## Attaching package: 'statnet.common'
```

```

## The following objects are masked from 'package:base':
##
## attr, order, replace

## Loading required package: network

##
## 'network' 1.20.0 (2026-02-06), part of the Statnet Project
## * 'news(package="network")' for changes since last version
## * 'citation("network")' for citation information
## * 'https://statnet.org' for help, support, and other information

## sna: Tools for Social Network Analysis
## Version 2.8 created on 2024-09-07.
## copyright (c) 2005, Carter T. Butts, University of California-Irvine
## For citation information, type citation("sna").
## Type help(package="sna") to get started.

library(parallel)

source("printSiena.R")

# Load data
f1 <- as.matrix(read.csv("Glasgow/f1.csv", header = FALSE))
f2 <- as.matrix(read.csv("Glasgow/f2.csv", header = FALSE))
f3 <- as.matrix(read.csv("Glasgow/f3.csv", header = FALSE))

demographic <- read.csv("Glasgow/demographic.csv", header = TRUE)
alcohol <- as.matrix(read.csv("Glasgow/alcohol.csv", header = TRUE))
logdistance <- as.matrix(read.csv("Glasgow/logdistance.csv", header = FALSE))

# Create Siena dependent network object
friendship <- sienaDependent(array(c(f1, f2, f3), dim = c(129, 129, 3)))

# Covariates
gender <- coCovar(demographic$gender)
age <- coCovar(demographic$age)
alcoholCov <- varCovar(alcohol)
distance <- coDyadCovar(logdistance)

# Create Siena data object
mydata <- sienaDataCreate(friendship, gender, age, alcoholCov, distance)

# Start model specification
myeff <- getEffects(mydata)

# Basic endogenous controls
myeff <- includeEffects(myeff, recip, transTrip, cycle3, name = "friendship")

## effectNumber effectName shortName include fix
## 1 18 reciprocity recip TRUE FALSE
## 2 24 transitive triplets transTrip TRUE FALSE
## 3 44 3-cycles cycle3 TRUE FALSE

```

```
## test initialValue parm
## 1 FALSE 0 0
## 2 FALSE 0 0
## 3 FALSE 0 0
```

```
# Basic exogenous controls: gender
myeff <- includeEffects(myeff, egoX, altX, sameX,
                        interaction1 = "gender",
                        name = "friendship")
```

```
## effectNumber effectName shortName include fix test
## 1 298 gender alter altX TRUE FALSE FALSE
## 2 313 gender ego egoX TRUE FALSE FALSE
## 3 369 same gender sameX TRUE FALSE FALSE
## initialValue parm
## 1 0 0
## 2 0 0
## 3 0 0
```

```
# Basic exogenous controls: age
myeff <- includeEffects(myeff, egoX, altX, simX,
                        interaction1 = "age",
                        name = "friendship")
```

```
## effectNumber effectName shortName include fix test
## 1 482 age alter altX TRUE FALSE FALSE
## 2 497 age ego egoX TRUE FALSE FALSE
## 3 543 age similarity simX TRUE FALSE FALSE
## initialValue parm
## 1 0 0
## 2 0 0
## 3 0 0
```

```
# Hypothesis i:
myeff <- includeEffects(myeff, inPop,
                        name = "friendship")
```

```
## effectNumber effectName shortName include
## 1 81 indegree - popularity inPop TRUE
## fix test initialValue parm
## 1 FALSE FALSE 0 0
```

```
# Hypothesis ii:
myeff <- includeEffects(myeff, egoX, altX, simX,
                        interaction1 = "alcoholCov",
                        name = "friendship")
```

```
## effectNumber effectName shortName include
## 1 666 alcoholCov alter altX TRUE
## 2 681 alcoholCov ego egoX TRUE
## 3 727 alcoholCov similarity simX TRUE
```

```
##   fix   test   initialValue   parm
## 1 FALSE FALSE           0     0
## 2 FALSE FALSE           0     0
## 3 FALSE FALSE           0     0
```

```
# Hypothesis iii:
myeff <- includeEffects(myeff, X,
                        interaction1 = "distance",
                        name = "friendship")
```

```
##   effectNumber effectName shortName include fix   test
## 1 265          distance    X          TRUE  FALSE FALSE
##   initialValue parm
## 1           0     0
```

```
myeff
```

```
##   effectName                                     include fix   test
## 1 constant friendship rate (period 1) TRUE      FALSE FALSE
## 2 constant friendship rate (period 2) TRUE      FALSE FALSE
## 3 outdegree (density)                     TRUE      FALSE FALSE
## 4 reciprocity                             TRUE      FALSE FALSE
## 5 transitive triplets                     TRUE      FALSE FALSE
## 6 3-cycles                               TRUE      FALSE FALSE
## 7 indegree - popularity                   TRUE      FALSE FALSE
## 8 distance                               TRUE      FALSE FALSE
## 9 gender alter                           TRUE      FALSE FALSE
## 10 gender ego                            TRUE      FALSE FALSE
## 11 same gender                           TRUE      FALSE FALSE
## 12 age alter                             TRUE      FALSE FALSE
## 13 age ego                              TRUE      FALSE FALSE
## 14 age similarity                         TRUE      FALSE FALSE
## 15 alcoholCov alter                      TRUE      FALSE FALSE
## 16 alcoholCov ego                       TRUE      FALSE FALSE
## 17 alcoholCov similarity                 TRUE      FALSE FALSE
##   initialValue parm
## 1      7.45424    0
## 2      6.82927    0
## 3     -1.61299    0
## 4      0.00000    0
## 5      0.00000    0
## 6      0.00000    0
## 7      0.00000    0
## 8      0.00000    0
## 9      0.00000    0
## 10     0.00000    0
## 11     0.00000    0
## 12     0.00000    0
## 13     0.00000    0
## 14     0.00000    0
## 15     0.00000    0
## 16     0.00000    0
## 17     0.00000    0
```

Here, we specify a SAOM where the friendship network is the dependent variable and evolves over time. For the endogenous variables in the network, we include effects for reciprocity, transitive triplets and three-cycles. Three-cycles are included alongside transitive triplets to separate hierarchical transitivity from cyclic closure patterns. We also include the exogenous covariates such as gender and age. To test the hypotheses, for hypothesis i we include the indegree popularity effect to see whether students tend to form ties with more popular students. For hypothesis ii, we include the variable of alcohol similarity to test whether students tend to form ties with others who have similar alcohol consumption levels. The ego and alter effects for alcohol are included as controls to account for the possibility that students with higher or lower alcohol consumption are systematically more or less socially active or popular independently of similarity. For hypothesis iii, we use the covariate for log-distance to test whether the geographical proximity increases the likelihood of students forming a tie.

Task 3.1.3

We estimate the model:

```
# Estimate the model
myAlgorithm <- set_algorithm_saom(seed = 1908, n3 = 3000, nsub = 4)

modelEv <- siena(data = mydata, effects = myeff, returnDeps = TRUE,
                 control_algo = myAlgorithm)
```

siena will create/use an output file mydata_out.txt .

```
# Results
summary(modelEv)
```

```
## Estimates, standard errors and convergence t-ratios
##
##
##
##
## Rate parameters:
## 0.1      Rate parameter period 1 11.0285 ( 1.0278 )
## 0.2      Rate parameter period 2  9.0751 ( 0.8272 )
##
## Other parameters:
## 1. eval outdegree (density)    -3.0771 ( 0.1286 )    0.0829
## 2. eval reciprocity           2.2154 ( 0.1038 )    0.0802
## 3. eval transitive triplets    0.6900 ( 0.0441 )    0.0788
## 4. eval 3-cycles              -0.4677 ( 0.0858 )    0.0708
## 5. eval indegree - popularity -0.0811 ( 0.0194 )    0.0707
## 6. eval distance              -0.1573 ( 0.0454 )    0.0032
## 7. eval gender alter          -0.1682 ( 0.1041 )   -0.0206
## 8. eval gender ego            0.1351 ( 0.1064 )   -0.0272
## 9. eval same gender           0.9000 ( 0.1004 )    0.0886
## 10. eval age alter            0.1171 ( 0.1166 )   -0.0131
## 11. eval age ego              -0.2146 ( 0.1130 )    0.0735
## 12. eval age similarity        0.0391 ( 0.1855 )    0.0116
## 13. eval alcoholCov alter      0.0299 ( 0.0319 )   -0.0433
## 14. eval alcoholCov ego       -0.0221 ( 0.0308 )   -0.0311
## 15. eval alcoholCov similarity 0.6376 ( 0.1628 )    0.0150
```

```

##
## Overall maximum convergence ratio:    0.1646
##
##
## Total of 4999 iteration steps.
##
## Covariance matrix of estimates (correlations below diagonal)
##
##      0.017      -0.003      0.000      0.001      -0.001      0.000      0.001      -0.001
##     -0.207      0.011      0.002     -0.005      0.000      0.000      0.000      0.000
##     -0.013      0.338      0.002     -0.003      0.000      0.000      0.000      0.000
##      0.083     -0.586     -0.775      0.007      0.000      0.000     -0.001      0.000
##     -0.578      0.035     -0.300      0.045      0.000      0.000      0.000      0.000
##     -0.044      0.043     -0.002      0.013      0.104      0.002      0.000      0.000
##      0.042     -0.001      0.025     -0.091      0.150     -0.049      0.011     -0.001
##     -0.095     -0.019     -0.054      0.104     -0.070      0.016     -0.736      0.000
##     -0.707     -0.033     -0.031      0.040     -0.003      0.011     -0.150      0.000
##     -0.024      0.089      0.045     -0.023      0.003      0.026      0.044     -0.001
##      0.057     -0.047      0.007     -0.009      0.009     -0.043      0.029     -0.001
##     -0.006      0.042      0.021     -0.050     -0.033     -0.025      0.001     -0.001
##      0.044     -0.010     -0.027      0.072     -0.072      0.024     -0.151      0.000
##     -0.006      0.020      0.051     -0.053      0.029      0.019     -0.013     -0.001
##     -0.094     -0.042     -0.050      0.030      0.037     -0.064     -0.036      0.000
##
## Derivative matrix of expected statistics X by parameters:
##
##      1126.520      741.809      2556.184      688.129      9296.291     -149.708       3.126      15.9
##       439.632      505.664      1373.254      459.019      3808.195     -83.322       5.259       3.0
##      1923.436     1594.709      9555.816     2802.385     19282.647    -370.074      34.078     43.8
##       817.975      856.176      4348.124     1464.756      8292.160    -173.755      19.548       7.1
##      5528.958     3671.947     15430.249     4131.093     52014.958    -887.321     -72.485     20.8
##      -192.291     -157.853     -633.490     -190.641     -1895.483      517.839       5.994       3.4
##       -15.641        6.133         3.624        13.426     -381.889      -4.440     271.946     233.0
##        13.314         1.425        28.154         1.076      -60.391     -10.610     233.991     280.4
##      1033.281      691.892      2263.395      601.200      8478.976    -136.117      -5.045     -5.5
##       -7.710     -20.232     -84.205     -31.463     -110.562       1.060     -8.273     -7.1
##      -25.097     -6.215     -52.213     -11.391     -226.478       7.679       0.260     -1.3
##       12.787       9.894       55.129       15.482      127.932       0.462       3.458       4.0
##        5.179      -5.418       14.747      -14.521     -65.495     -44.209     162.358     146.8
##       16.216       36.166       47.660       28.909     -32.443     -44.144     144.708     160.0
##        61.141       48.916      144.500       43.181      488.652       1.463      -4.486     -3.5
##
## Covariance matrix of X (correlations below diagonal):
##
##      1502.580      1142.573      5248.807      1494.246      13731.716     -261.519       27.390      48.5
##        0.856      1187.078      4892.834      1549.563      10791.347     -242.959       29.967      28.0
##        0.749        0.786     32660.923      9896.141      56931.841    -1209.108      216.558     241.4
##        0.684        0.798         0.972      3174.889      16363.392     -369.478       73.074      64.0
##        0.952        0.842         0.847         0.781     138386.640    -2668.100       21.289     227.0
##       -0.293     -0.306     -0.290     -0.284       -0.311      531.264     -29.398     -28.4
##        0.038        0.047         0.064         0.069         0.003       -0.068     348.890     333.4
##        0.064        0.042         0.068         0.059         0.031       -0.063       0.911     383.7
##        0.956        0.827         0.687         0.619         0.903       -0.277       0.010       0.0
##        0.032        0.000         -0.014         -0.017         0.017       -0.019       0.005       0.0

```

##	-0.072	-0.031	-0.054	-0.038	-0.078	0.014	0.013	-0.0
##	0.073	0.068	0.056	0.048	0.077	0.007	0.015	0.0
##	0.062	0.064	0.050	0.043	0.050	-0.084	0.338	0.3
##	0.006	0.041	0.000	0.010	-0.007	-0.064	0.314	0.3
##	0.304	0.297	0.236	0.231	0.284	-0.025	-0.021	-0.0

```
# Check convergence
modelEv$tconv.max
```

```
##          [,1]
## [1,] 0.1646112
```

Comments: The SAOM was estimated using a stochastic approximation algorithm with 3000 phase 3 simulations. The overall maximum convergence ratio is 0.1646, which is below 0.25 and therefore indicates satisfactory convergence. Also, good convergence is indicated by the t-ratios being close to zero.

Now let's look at the parameters: The estimated parameters in the model indicate some important patterns in the evolution of the friendship network. Statistical significance is assessed by computing the t-ratio, defined as the estimated coefficient divided by its standard error. A t-ratio with absolute value greater than 2 is generally considered statistically significant.

The estimate for density (outdegree) is -3.08 (with a standard error of 0.129). This is a significant negative outdegree density, indicating that the networks are sparse and therefore students tend to not form many ties. The estimate for reciprocity is 2.22 (with a standard error of 0.104). This is a strong positive estimator, which indicated that students strongly tend to reciprocate friendships. The estimate for transitive triplets is 0.690 (with a standard error of 0.044). This indicates a strong triadic closure, which means that students are more likely to be tied to friends of their friends. The estimate for 3-cycles is -0.468 (with a standard error of 0.086), which indicates that the network is more hierarchical than cyclic. The gender homophily effect is strongly positive, since the estimate is 0.900 (with a standard error of 0.100), indicating that students strongly prefer same-gender friendships.

Now, looking at the three hypotheses: The indegree popularity estimate is -0.081 with a standard error of 0.019. So the indegree effect is negative and statistically significant, which suggests that students are less likely to form ties with students who are popular. This means that Hypothesis i is not supported by the data. The alcohol similarity estimate is 0.638 with a standard error of 0.163. This shows that the alcohol similarity effect is positive and significant, indicating that students tend to form friendships with others that have similar alcohol consumption levels. This supports Hypothesis ii. The distance estimate is -0.157 with a standard error of 0.045. This shows a significant negative effect, indicating that students are more likely to form friendships when they live closer to each other, supporting Hypothesis iii.

The remaining control effects for age and gender are significant, confirming that they are important co-founders to account for in the model.

Task 3.1.4

We evaluate the goodness of fit of the estimated model using the following auxiliary statistics: indegree distribution, outdegree distribution, triad census and geodesic distance.

```
# Geodesic distance function
GeodesicDistribution <- function(
  i, data, sims, period, groupName,
  varName, levls = c(1:5, Inf), cumulative = TRUE
) {
  x <- networkExtraction(i, data, sims, period, groupName, varName)
```

```

require(sna)
a <- sna::geodist(symmetrize(x))$gdist

if (cumulative) {
  gdi <- sapply(levls, function(i) sum(a <= i))
} else {
  gdi <- sapply(levls, function(i) sum(a == i))
}

names(gdi) <- as.character(levls)
gdi
}

# Use parallel computation
cl <- makeCluster(4)

# Indegree distribution
gofEvId <- test_gof(
  modelEv,
  verbose = FALSE,
  varName = "friendship", IndegreeDistribution,
  cluster = cl
)

# Outdegree distribution
gofEvOd <- test_gof(
  modelEv,
  verbose = FALSE,
  varName = "friendship", OutdegreeDistribution,
  cluster = cl
)

# Triad census
gofEvTC <- test_gof(
  modelEv,
  verbose = FALSE,
  varName = "friendship", TriadCensus,
  cluster = cl
)

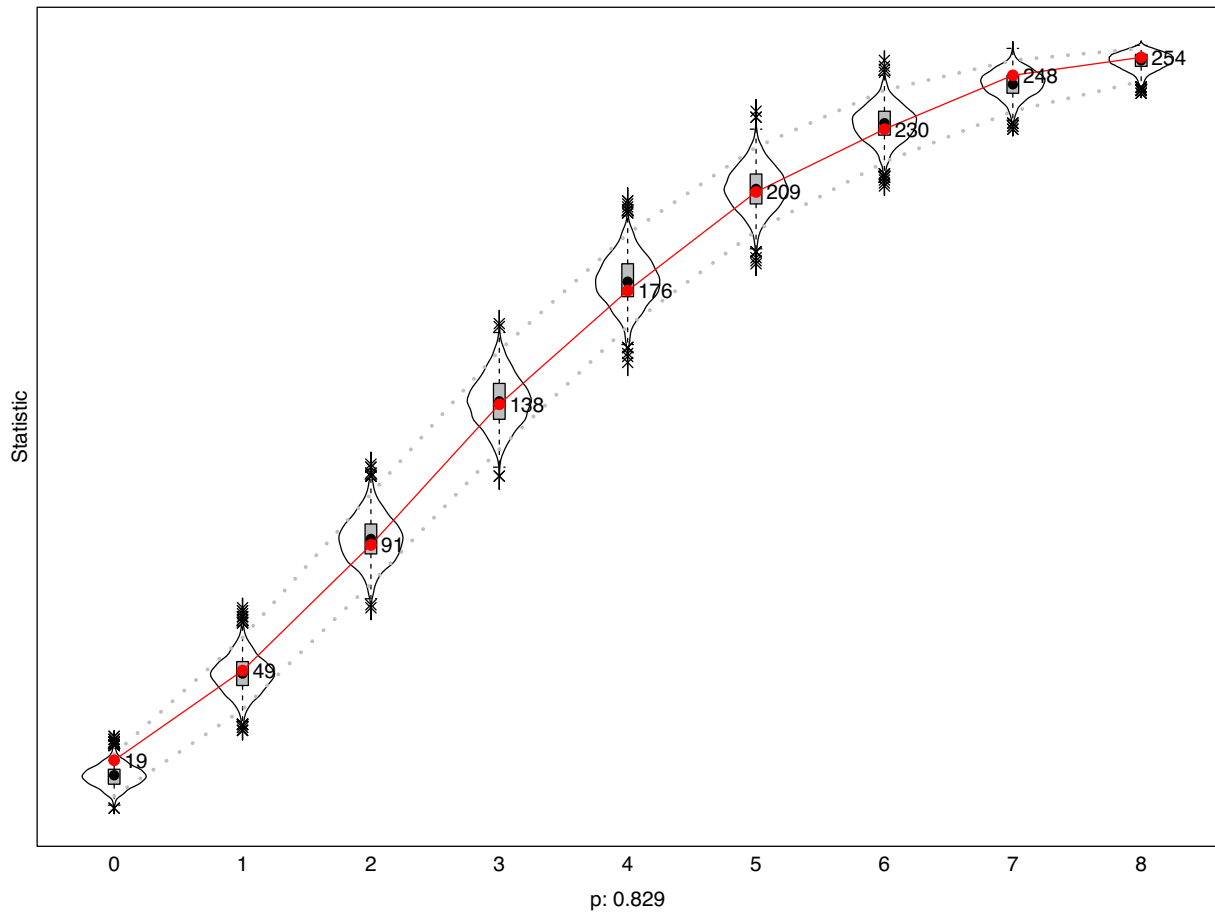
# Geodesic distance
gofEvGD <- test_gof(
  modelEv,
  verbose = FALSE,
  varName = "friendship", GeodesicDistribution
)

stopCluster(cl)

# Plots
plot(gofEvId)

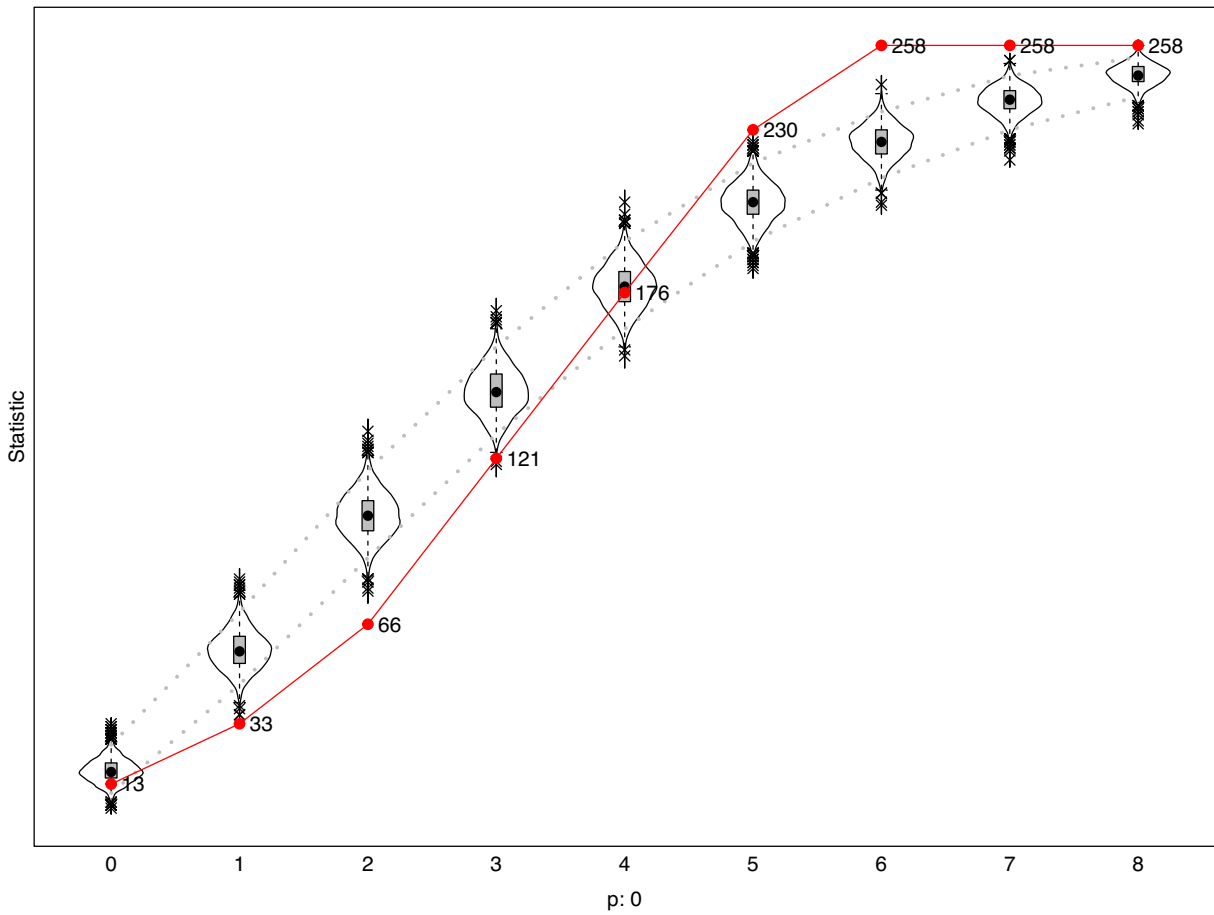
```

Goodness of Fit of IndegreeDistribution



```
plot(gofEv0d)
```

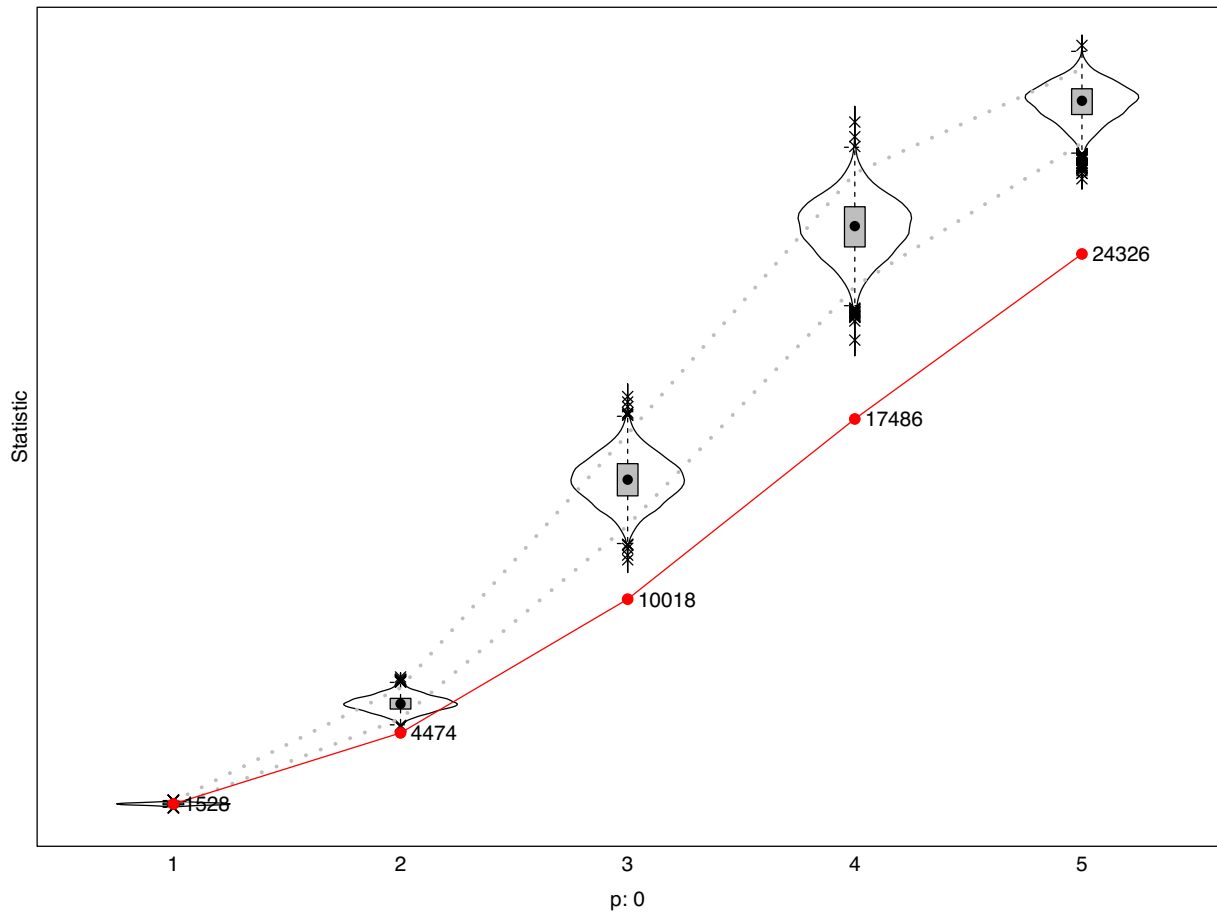
Goodness of Fit of OutdegreeDistribution



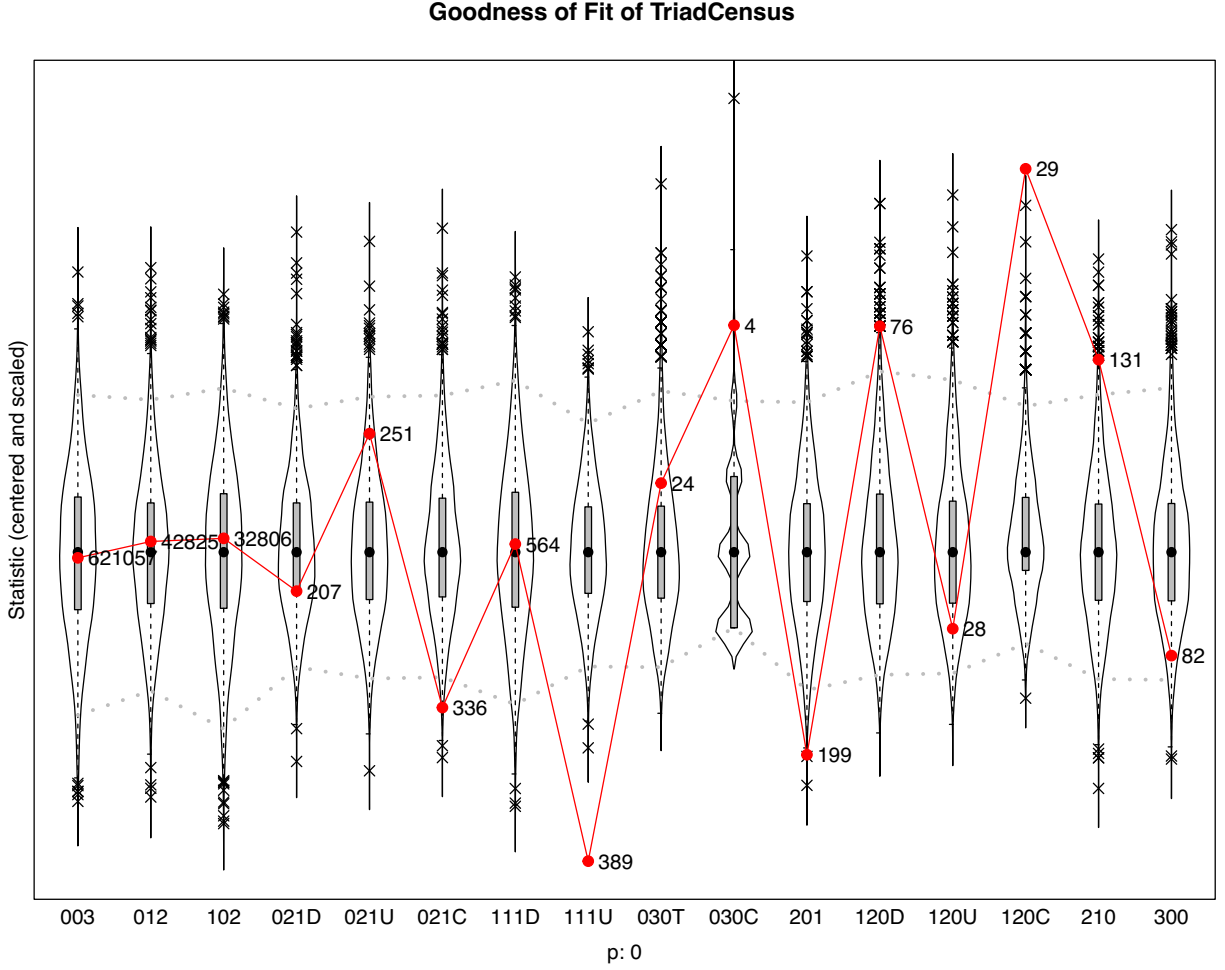
```
plot(gofEvGD)
```

```
## Note: some statistics are not plotted because their variance is 0.  
## This holds for the statistic: Inf.
```

Goodness of Fit of GeodesicDistribution



```
plot(gofEvTC, center = TRUE, scale = TRUE)
```



Comments: Looking at the plot for the goodness of fit of the indegree distribution, we observe that the red line, i.e. the observed network is almost entirely inside the grey distributions, which are the simulated networks, across all indegree levels. This shows that the model fits the indegree distribution very well and successfully captures the distribution of the popularity in the network. Also, the high p-value of 0.829 indicates that there is no significant deviation between the observed distributions and the simulated ones.

In contrast, in the plot for the goodness of fit of the outdegree distribution we see a poor model fit. The red line (observed network) is mostly outside the range of grey distributions (simulated distributions). This suggests that the model doesn't adequately represent the distribution of outgoing ties among students. This is likely due to the lack of an outdegree effect in the model. The p-value of 0 further supports these observations and indicates a significant discrepancy between the observed and simulated distributions and therefore we have strong evidence of model misfit for the outdegree distribution.

In the plot for the goodness of fit of the geodesic distance distribution we see a poor model fit. The red line (observed network) is mostly outside the range of grey distributions (simulated distributions) and the observed values are consistently lower than the simulated values. More specifically, the observed network seems to have shorter path lengths between actors compared to the simulated one. This suggests that the model doesn't adequately represent the geodesic distance distribution. This is likely due to the lack of a connectivity effect which would capture how students connect across different social groups. The p-value of 0 further supports these observations and indicates a significant discrepancy between the observed and simulated distributions and therefore we have strong evidence of model misfit for the geodesic distance distribution.

The plot for goodness of fit of the triad census also shows a poor fit, since several observed triad counts are

outside the range of the simulated distributions. The p-value of 0 further supports that the model doesn't accurately represent the triad census distribution and doesn't fully capture all the aspects of the local network structure, even though it includes transitive triplets and three-cycles.

Overall, we see that although the model successfully captures the distribution of popularity, it fails to reproduce the outdegree, the geodesic distance and the triad census distributions. This suggests that additional effects may be required to capture the full structure of the network.

Task 3.1.5

To evaluate the hypotheses i-iii, we examine the estimated coefficients and their significance. The statistical significance can be measured by dividing the estimated coefficient by its standard error.

For Hypothesis i, we were testing whether students tend to be friends with popular pupils. Our results showed that the indegree popularity effect is negative and statistically significant, indicating that students don't tend to form ties with pupils who are more popular. This suggests that students tend to avoid highly popular pupils, or that the formation of ties is not driven by popularity. Therefore, Hypothesis i is not supported by the data.

For Hypothesis ii, we were testing whether students tend to be friends with pupils with similar alcohol consumption levels. To do this, we looked at the alcohol similarity effect. The estimate for this coefficient is positive and statistically significant, indicating that students are more likely to form ties with other students whose alcohol consumption levels are similar to theirs. This shows evidence for homophily based on alcohol consumption and therefore Hypothesis ii is supported by the data,

For Hypothesis iii, we were testing whether student tend to be friends with students who live in the same neighbourhood. For this, we examined the effect of log-distance between student's homes. From our results, the estimated coefficient is negative and statistically significant, indicating that that students are more likely to form ties with pupils whose distance between their homes is shorter. This supports the idea that physical proximity increases the probability of forming a friendship and therefore Hypothesis iii is supported by the data.

Looking at the magnitude of the effects, the alcohol similarity effect (0.638) is notably larger than the distance effect (-0.157), suggesting that behavioral homophily plays a stronger role in friendship formation than geographical proximity.

Overall, the results show that the formation of ties between students is driven by similarity in behaviours such as alcohol consumption as well as geographical proximity, rather than popularity. Students showed a tendency to befriend others who had similar alcohol consumption levels and lived closer to them, while there is no evidence to support the hypothesis that students are more likely to form ties with more popular pupils. In general, this indicates a significance of homophily in the formation of student friendships.

Task 3.2.1

We investigate the co-evolution of friendship and alcohol consumption, by using the model specification developed in (1).

```
# Dependent variables
friendship <- sienaDependent(array(c(f1, f2, f3), dim = c(129, 129, 3)))
alcoholDep <- sienaDependent(alcohol, type = "behavior")

# Covariates
gender <- coCovar(demographic$gender)
age <- coCovar(demographic$age)
distance <- coDyadCovar(logdistance)
```

```

# Siena data object
mydataCoev <- sienaDataCreate(friendship, alcoholDep, gender, age, distance)

# Effects object
myeffCoev <- getEffects(mydataCoev)

## Selection part: friendship evolution -----

# Basic endogenous controls
myeffCoev <- includeEffects(myeffCoev, recip, transTrip, cycle3,
                             name = "friendship")

```

```

##   effectNumber effectName      shortName include fix
## 1 18           reciprocity      recip    TRUE  FALSE
## 2 24           transitive triplets transTrip TRUE  FALSE
## 3 44           3-cycles          cycle3   TRUE  FALSE
##   test  initialValue parm
## 1 FALSE          0    0
## 2 FALSE          0    0
## 3 FALSE          0    0

```

```

# Gender controls
myeffCoev <- includeEffects(myeffCoev, egoX, altX, sameX,
                             interaction1 = "gender",
                             name = "friendship")

```

```

##   effectNumber effectName  shortName include fix  test
## 1 298          gender alter altX      TRUE  FALSE FALSE
## 2 313          gender ego   egoX      TRUE  FALSE FALSE
## 3 369          same gender sameX      TRUE  FALSE FALSE
##   initialValue parm
## 1          0    0
## 2          0    0
## 3          0    0

```

```

# Age controls
myeffCoev <- includeEffects(myeffCoev, egoX, altX, simX,
                             interaction1 = "age",
                             name = "friendship")

```

```

##   effectNumber effectName      shortName include fix  test
## 1 482          age alter      altX      TRUE  FALSE FALSE
## 2 497          age ego        egoX      TRUE  FALSE FALSE
## 3 543          age similarity simX      TRUE  FALSE FALSE
##   initialValue parm
## 1          0    0
## 2          0    0
## 3          0    0

```

```

# Hypothesis i:
myeffCoev <- includeEffects(myeffCoev, inPop,
                           name = "friendship")

##   effectNumber effectName          shortName include
## 1 81           indegree - popularity inPop      TRUE
##   fix   test  initialValue parm
## 1 FALSE FALSE           0    0

# Hypothesis ii:
myeffCoev <- includeEffects(myeffCoev, egoX, altX, simX,
                           interaction1 = "alcoholDep",
                           name = "friendship")

##   effectNumber effectName          shortName include
## 1 666           alcoholDep alter      altX      TRUE
## 2 681           alcoholDep ego        egoX      TRUE
## 3 727           alcoholDep similarity simX      TRUE
##   fix   test  initialValue parm
## 1 FALSE FALSE           0    0
## 2 FALSE FALSE           0    0
## 3 FALSE FALSE           0    0

# Hypothesis iii:
myeffCoev <- includeEffects(myeffCoev, X,
                           interaction1 = "distance",
                           name = "friendship")

##   effectNumber effectName shortName include fix   test
## 1 265           distance  X           TRUE   FALSE FALSE
##   initialValue parm
## 1           0    0

# Basic behavioural controls
myeffCoev <- includeEffects(myeffCoev, linear, quadratic,
                           name = "alcoholDep")

##   effectNumber effectName          shortName include
## 1 910           alcoholDep linear shape linear      TRUE
##   fix   test  initialValue parm
## 1 FALSE FALSE    0.34704    0

# Hypothesis iv:
myeffCoev <- includeEffects(myeffCoev, indeg,
                           name = "alcoholDep",
                           interaction1 = "friendship")

##   effectNumber effectName          shortName include fix
## 1 980           alcoholDep indegree indeg      TRUE   FALSE
##   test  initialValue parm
## 1 FALSE           0    0

```

Hypothesis v:

```
myeffCoev <- includeEffects(myeffCoev, avAlt,
                             name = "alcoholDep",
                             interaction1 = "friendship")
```

```
##   effectNumber effectName          shortName include
## 1 999          alcoholDep average alter avAlt     TRUE
##   fix   test  initialValue parm
## 1 FALSE FALSE           0     0
```

myeffCoev

```
##   name          effectName          include
## 1 friendship constant friendship rate (period 1) TRUE
## 2 friendship constant friendship rate (period 2) TRUE
## 3 friendship outdegree (density)      TRUE
## 4 friendship reciprocity              TRUE
## 5 friendship transitive triplets       TRUE
## 6 friendship 3-cycles                  TRUE
## 7 friendship indegree - popularity     TRUE
## 8 friendship distance                  TRUE
## 9 friendship gender alter              TRUE
## 10 friendship gender ego               TRUE
## 11 friendship same gender              TRUE
## 12 friendship age alter                 TRUE
## 13 friendship age ego                   TRUE
## 14 friendship age similarity            TRUE
## 15 friendship alcoholDep alter         TRUE
## 16 friendship alcoholDep ego           TRUE
## 17 friendship alcoholDep similarity    TRUE
## 18 alcoholDep rate alcoholDep (period 1) TRUE
## 19 alcoholDep rate alcoholDep (period 2) TRUE
## 20 alcoholDep alcoholDep linear shape  TRUE
## 21 alcoholDep alcoholDep quadratic shape TRUE
## 22 alcoholDep alcoholDep indegree      TRUE
## 23 alcoholDep alcoholDep average alter  TRUE
##   fix   test  initialValue parm
## 1 FALSE FALSE    7.45424    0
## 2 FALSE FALSE    6.82927    0
## 3 FALSE FALSE   -1.61299    0
## 4 FALSE FALSE    0.00000    0
## 5 FALSE FALSE    0.00000    0
## 6 FALSE FALSE    0.00000    0
## 7 FALSE FALSE    0.00000    0
## 8 FALSE FALSE    0.00000    0
## 9 FALSE FALSE    0.00000    0
## 10 FALSE FALSE   0.00000    0
## 11 FALSE FALSE   0.00000    0
## 12 FALSE FALSE   0.00000    0
## 13 FALSE FALSE   0.00000    0
## 14 FALSE FALSE   0.00000    0
## 15 FALSE FALSE   0.00000    0
## 16 FALSE FALSE   0.00000    0
```

```
## 17 FALSE FALSE      0.00000  0
## 18 FALSE FALSE      0.91000  0
## 19 FALSE FALSE      1.09314  0
## 20 FALSE FALSE      0.34704  0
## 21 FALSE FALSE      0.00000  0
## 22 FALSE FALSE      0.00000  0
## 23 FALSE FALSE      0.00000  0
```

For the behaviour part of the model, we include linear and quadratic shape effects as basic controls for the distribution of alcohol consumption. The linear effect captures the overall trend in alcohol consumption, while the quadratic effect captures whether students tend to avoid extreme values. To test hypothesis iv, we include the indegree effect on alcohol consumption, which captures whether students who receive many friendship nominations tend to increase their alcohol consumption level. To test hypothesis v, we include the average alter effect, which captures whether students tend to adjust their alcohol consumption towards the average level of their friends.

Task 3.2.2

We estimate the model:

```
# Estimation algorithm
myAlgorithm <- set_algorithm_saom(seed = 1908, n3 = 3000, nsub = 4)

# Estimate co-evolution model
modelCoev <- siena(data = mydataCoev,
  effects = myeffCoev,
  returnDeps = TRUE,
  control_algo = myAlgorithm)
```

siena will create/use an output file mydataCoev_out.txt .

```
# Results
summary(modelCoev)
```

```
## Estimates, standard errors and convergence t-ratios
##
##
##
##
## Network Dynamics
## 1. rate constant friendship rate (period 1) 10.8773 ( 1.0002 ) -0.0525
## 2. rate constant friendship rate (period 2) 8.9740 ( 0.8321 ) 0.0456
## 3. eval outdegree (density) -3.1092 ( 0.1322 ) -0.0401
## 4. eval reciprocity 2.1954 ( 0.1139 ) -0.0302
## 5. eval transitive triplets 0.6815 ( 0.0451 ) -0.0746
## 6. eval 3-cycles -0.4604 ( 0.0909 ) -0.0700
## 7. eval indegree - popularity -0.0792 ( 0.0201 ) -0.0406
## 8. eval distance -0.1621 ( 0.0458 ) 0.0060
## 9. eval gender alter -0.1736 ( 0.1081 ) -0.0261
## 10. eval gender ego 0.1444 ( 0.1072 ) -0.0066
## 11. eval same gender 0.8985 ( 0.0982 ) -0.0268
## 12. eval age alter 0.1175 ( 0.1166 ) -0.0318
## 13. eval age ego -0.2221 ( 0.1130 ) 0.0100
```

```
## 14. eval age similarity          0.0477 ( 0.1970 )    0.0356
## 15. eval alcoholDep alter       0.0462 ( 0.0497 )    0.0064
## 16. eval alcoholDep ego        -0.0431 ( 0.0481 )   -0.0024
## 17. eval alcoholDep similarity  1.2344 ( 0.3760 )   -0.0125
```

```
##
```

```
## Behavior Dynamics
```

```
## 18. rate rate alcoholDep (period 1)  1.5555 ( 0.2735 )   -0.0121
## 19. rate rate alcoholDep (period 2)  2.1851 ( 0.3526 )   -0.0224
## 20. eval alcoholDep linear shape     0.1293 ( 0.2574 )   -0.0096
## 21. eval alcoholDep quadratic shape  -0.4258 ( 0.1250 )   -0.0334
## 22. eval alcoholDep indegree        0.0741 ( 0.0685 )   -0.0053
## 23. eval alcoholDep average alter    1.0316 ( 0.3671 )   -0.0170
```

```
##
```

```
## Overall maximum convergence ratio: 0.1554
```

```
##
```

```
##
```

```
## Total of 5815 iteration steps.
```

```
##
```

```
## Covariance matrix of estimates (correlations below diagonal)
```

```
##
```

##	1.000	-0.058	0.020	-0.001	-0.007	0.009	0.000	0.000
##	-0.069	0.692	0.017	-0.004	-0.003	0.003	-0.001	0.000
##	0.148	0.157	0.017	-0.003	0.000	0.001	-0.002	0.000
##	-0.006	-0.037	-0.190	0.013	0.002	-0.006	0.000	0.000
##	-0.165	-0.079	-0.014	0.380	0.002	-0.003	0.000	0.000
##	0.094	0.046	0.063	-0.599	-0.792	0.008	0.000	0.000
##	0.020	-0.031	-0.638	0.084	-0.247	0.039	0.000	0.000
##	0.035	0.025	-0.034	-0.024	-0.072	0.071	0.135	0.000
##	0.073	0.049	0.072	0.000	0.017	-0.087	0.098	0.000
##	-0.099	-0.020	-0.124	-0.023	-0.026	0.088	-0.018	-0.000
##	-0.111	-0.096	-0.648	-0.141	-0.093	0.110	0.030	0.000
##	0.045	-0.033	-0.072	0.076	0.004	-0.012	0.032	0.000
##	-0.061	0.044	0.024	-0.081	0.034	-0.033	-0.002	0.000
##	0.002	-0.004	-0.013	0.003	0.000	-0.015	-0.034	-0.000
##	-0.010	-0.050	-0.039	0.054	-0.051	0.065	-0.041	0.000
##	0.038	0.023	0.012	0.009	0.112	-0.107	0.019	-0.000
##	-0.184	-0.126	-0.186	-0.050	0.046	-0.050	-0.026	-0.000
##	-0.032	-0.055	-0.020	-0.074	-0.080	0.084	0.010	-0.000
##	-0.045	0.069	-0.015	0.011	-0.055	0.031	0.048	0.000
##	0.010	-0.047	-0.012	0.001	0.039	-0.022	0.000	-0.000
##	0.004	-0.076	0.057	0.057	-0.034	0.036	0.005	0.000
##	0.018	0.083	0.013	-0.006	-0.027	0.011	-0.003	0.000
##	0.026	0.073	-0.025	-0.073	0.013	-0.015	-0.022	-0.000

```
##
```

```
## Derivative matrix of expected statistics X by parameters:
```

```
##
```

##	19.182	0.000	1.345	1.531	26.292	8.610	30.173	-0.000
##	0.000	20.442	-2.206	-0.573	9.709	3.184	-3.480	-0.000
##	142.780	124.776	1173.673	795.515	2962.116	833.695	10026.187	-187.500
##	15.385	5.925	436.761	498.276	1350.191	456.456	3735.563	-70.000
##	249.302	150.778	1845.756	1548.648	9512.848	2819.166	18755.603	-363.400
##	58.421	40.911	800.013	840.753	4322.030	1457.725	8098.324	-178.000
##	794.468	640.985	5647.447	3827.114	16853.871	4693.736	54274.356	-1058.000
##	-35.657	-23.131	-163.693	-121.700	-585.592	-168.844	-1756.888	519.000

```

##      -15.169      -9.787      -21.056      -0.559       8.989      15.176      -378.113      -6.2
##      -12.936      -0.754      -17.514      -13.676      -30.696      -8.755      -317.968      -7.4
##      146.112      121.346      1059.112      738.688      2653.057      738.599      9005.882      -173.5
##       4.018       1.196       2.196      -12.060      -4.032      -6.173       6.849      -0.9
##      -8.441      -2.564      -7.382       12.734      16.534      12.870      -60.055      -3.2
##      -0.987      -2.391       7.394       6.277      26.173       7.634      80.480       2.1
##     -17.705      28.219      27.334     -15.617     -37.769     -33.786      61.469     -42.3
##     -38.406      17.644     -62.029     -25.991     -236.970     -50.248     -736.754     -15.6
##      16.876      11.583      77.406      59.007      211.070      64.620      671.474      -3.5
##      -0.405       0.000      -2.802      -0.720       3.386      -1.425     -24.522      1.9
##       0.000      -6.187      -3.305      -1.930      -8.147      -3.242     -32.003     -1.4
##      -2.330       0.351      -7.284      -9.704     -42.616     -13.546     -69.369       6.8
##     -22.422     -18.354     -119.382     -67.378     -178.036     -51.454     -922.701     33.3
##      -8.991       0.689     -16.387     -40.006     -146.668     -62.198     -179.058     18.6
##      -5.547      -9.093     -17.437      -0.788      11.248       6.307     -126.198      7.4
##
## Covariance matrix of X (correlations below diagonal):
##
##      382.363      -0.730      231.257      121.110      915.624      262.584      2447.144      -56.7
##      -0.002      318.007      150.914       55.175      447.096      121.577      1454.441     -35.7
##       0.300       0.214     1558.208     1201.454     5674.755     1666.065     14500.149     -294.7
##       0.176       0.088       0.864     1241.170     5188.900     1661.016     11426.572     -244.4
##       0.254       0.136       0.780       0.799     33971.316     10439.530     61413.363    -1358.5
##       0.231       0.117       0.727       0.812       0.976     3367.933     18115.501     -411.7
##       0.325       0.212       0.955       0.843       0.866       0.811     148000.994    -3126.6
##      -0.125      -0.086      -0.321      -0.298      -0.316      -0.305      -0.349     542.6
##      -0.004      -0.048       0.013       0.044       0.088       0.096       0.001      -0.6
##       0.005      -0.036       0.039       0.042       0.092       0.086       0.031      -0.6
##       0.267       0.210       0.957       0.834       0.722       0.665       0.904     -0.3
##       0.017       0.004       0.013      -0.019       0.012       0.006       0.012       0.6
##      -0.046      -0.028      -0.084      -0.034      -0.058      -0.040      -0.087       0.6
##      -0.011      -0.006       0.061       0.048       0.049       0.044       0.065     -0.6
##      -0.115      -0.021      -0.168      -0.139      -0.140      -0.135      -0.174     -0.6
##      -0.105      -0.046      -0.218      -0.158      -0.168      -0.151      -0.216       0.6
##       0.115       0.030       0.308       0.298       0.269       0.255       0.299     -0.6
##       0.012       0.002       0.042       0.033       0.053       0.044       0.044     -0.6
##       0.002       0.004       0.017       0.004       0.000      -0.004       0.020     -0.6
##       0.000      -0.001       0.087       0.063       0.069       0.058       0.083     -0.6
##      -0.034      -0.063      -0.091      -0.047      -0.028      -0.022      -0.078       0.6
##       0.010       0.007       0.091       0.066       0.076       0.062       0.092     -0.6
##      -0.015      -0.085       0.013       0.035       0.032       0.034       0.008       0.6
##
# Check convergence
modelCoev$tconv.max

##      [,1]
## [1,] 0.155407

```

Comments: The co-evolution model was estimated using a stochastic approximation algorithm with 3000 phase 3 simulations. The overall maximum convergence ratio is 0.1711, which is below 0.25 and therefore indicates satisfactory convergence. Also, good convergence is indicated by the t-ratios being close to zero, suggesting that the estimation procedure reproduced the observed network behavioural statistics well.

Now let's look at the parameters: Overall, the network parameters appear to be quite consistent with the parameters in the friendship-only model. The outdegree parameter estimate is negative (-3.10, with

standard error 0.13), which indicates that the friendship network is sparse. The reciprocity parameter estimate is positive and statistically significant (2.19, with standard error 0.11), indicating that students tend to reciprocate their friendships. The transitive triplets estimate is also positive and significant (0.68, with standard error 0.04), indicating triadic closure, while the three-cycle effect estimate is negative and significant (-0.46, with standard error 0.08) and suggests a more hierarchical than cyclic structure.

The indegree popularity effect estimate is negative and significant (-0.08, with standard error 0.02), indicating that students are less likely to form ties with popular students. The distance effect estimate is also negative and significant (-0.16, with standard error 0.05), suggesting that students are more likely to form ties with others who live close to them. The gender effect is positive and significant (0.90 with standard error 0.10), indicating strong gender homophily. The alcohol similarity estimate is also positive and significant (1.20, with standard error 0.36), showing that students tend to select friends with similar alcohol consumption levels.

Now looking at the behaviour dynamics: The quadratic shape effect is negative and significant (-0.42 with standard error 0.14), suggesting that students are less likely to have extreme alcohol consumption levels, while the linear shape effect (0.16, with standard error 0.24) is not statistically significant, meaning there is no overall directional trend in alcohol consumption. The indegree effect on alcohol consumption is positive but not significant (0.07, with standard error 0.07), indicating that we have no clear evidence that more popular students drink more. In contrast, the average alter effect is positive and significant (1.03, with standard error 0.40), showing that we have evidence that students tend to adjust their alcohol consumption to that of their friends.

Task 3.2.3

We evaluate the goodness of fit of the estimated model using the following auxiliary statistics: indegree distribution, outdegree distribution, triad census, geodesic distance and behavioral distribution.

```
## Goodness of fit

# Geodesic distance function
GeodesicDistribution <- function(
  i, data, sims, period, groupName,
  varName, levls = c(1:5, Inf), cumulative = TRUE
) {
  x <- networkExtraction(i, data, sims, period, groupName, varName)
  require(sna)
  a <- sna::geodist(symmetrize(x))$gdist

  if (cumulative) {
    gdi <- sapply(levls, function(i) sum(a <= i))
  } else {
    gdi <- sapply(levls, function(i) sum(a == i))
  }

  names(gdi) <- as.character(levls)
  gdi
}

cl <- makeCluster(4)
clusterEvalQ(cl, library(RSiena))

## [[1]]
## [1] "RSiena"      "stats"      "graphics"   "grDevices"
```

```
## [5] "utils"      "datasets"   "methods"    "base"
##
## [[2]]
## [1] "RSiena"     "stats"      "graphics"    "grDevices"
## [5] "utils"      "datasets"   "methods"    "base"
##
## [[3]]
## [1] "RSiena"     "stats"      "graphics"    "grDevices"
## [5] "utils"      "datasets"   "methods"    "base"
##
## [[4]]
## [1] "RSiena"     "stats"      "graphics"    "grDevices"
## [5] "utils"      "datasets"   "methods"    "base"
```

```
clusterEvalQ(cl, library(sna))
```

```
## [[1]]
## [1] "sna"          "network"      "statnet.common"
## [4] "RSiena"       "stats"        "graphics"
## [7] "grDevices"    "utils"        "datasets"
## [10] "methods"      "base"
##
## [[2]]
## [1] "sna"          "network"      "statnet.common"
## [4] "RSiena"       "stats"        "graphics"
## [7] "grDevices"    "utils"        "datasets"
## [10] "methods"      "base"
##
## [[3]]
## [1] "sna"          "network"      "statnet.common"
## [4] "RSiena"       "stats"        "graphics"
## [7] "grDevices"    "utils"        "datasets"
## [10] "methods"      "base"
##
## [[4]]
## [1] "sna"          "network"      "statnet.common"
## [4] "RSiena"       "stats"        "graphics"
## [7] "grDevices"    "utils"        "datasets"
## [10] "methods"      "base"
```

```
clusterExport(cl, "GeodesicDistribution")
```

```
gofId <- test_gof(
  modelCoev,
  auxiliaryFunction = IndegreeDistribution,
  varName = "friendship",
  tested = FALSE,
  cluster = cl
)

gofOd <- test_gof(
  modelCoev,
  auxiliaryFunction = OutdegreeDistribution,
```

```

    varName = "friendship",
    tested = FALSE,
    cluster = cl
)

gofTC <- test_gof(
  modelCoev,
  auxiliaryFunction = TriadCensus,
  varName = "friendship",
  tested = FALSE,
  cluster = cl
)

gofGD <- test_gof(
  modelCoev,
  auxiliaryFunction = GeodesicDistribution,
  varName = "friendship",
  tested = FALSE
)

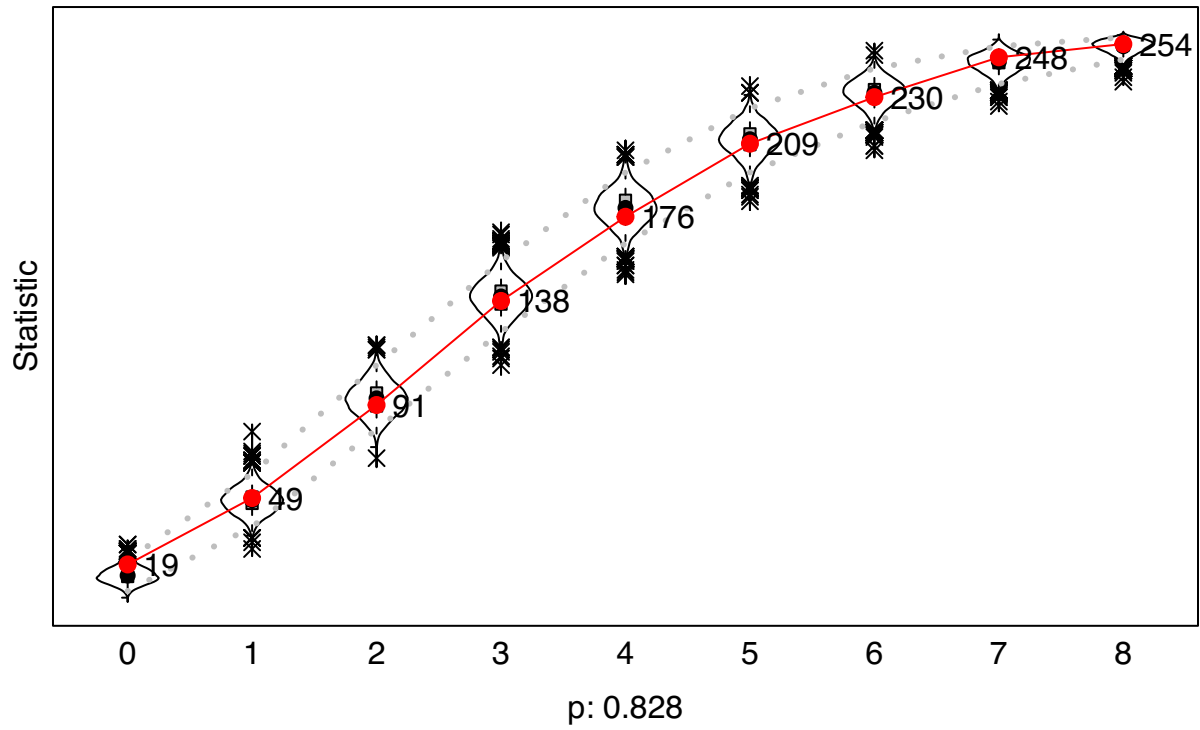
gofBeh <- test_gof(
  modelCoev,
  auxiliaryFunction = BehaviorDistribution,
  varName = "alcoholDep",
  tested = FALSE,
  cluster = cl
)

stopCluster(cl)

# Plots
plot(gofId)

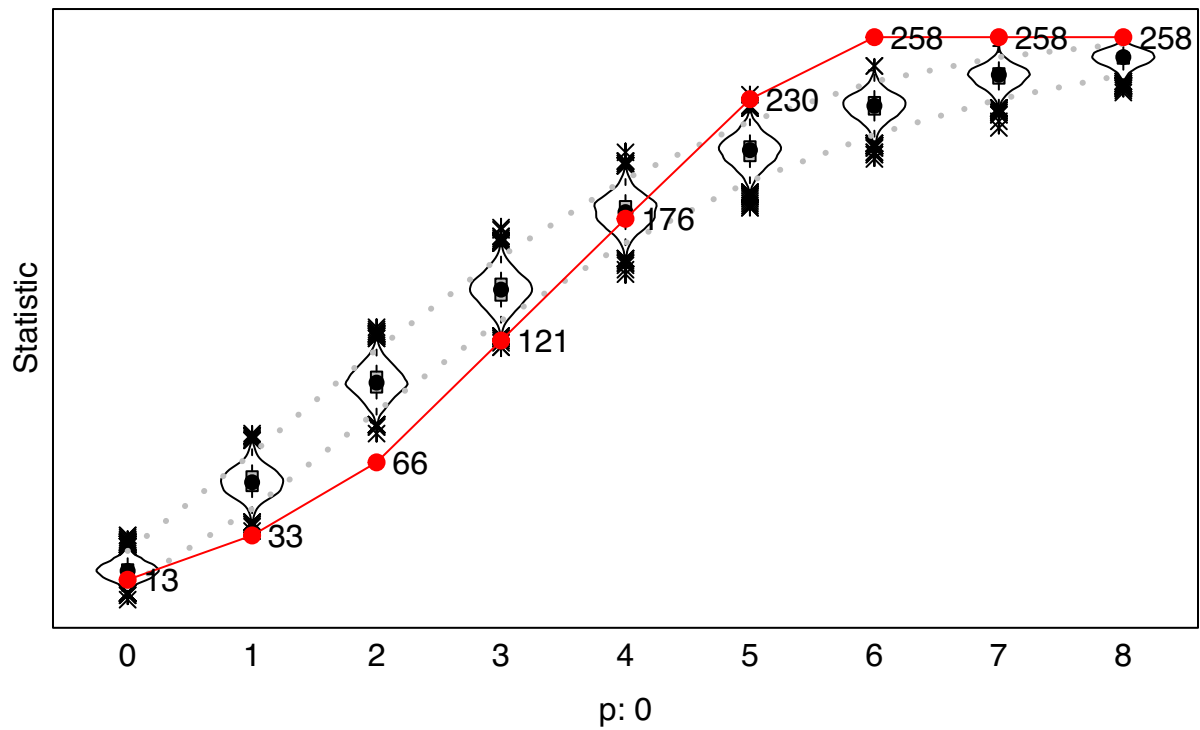
```

Goodness of Fit of IndegreeDistribution



```
plot(gof0d)
```

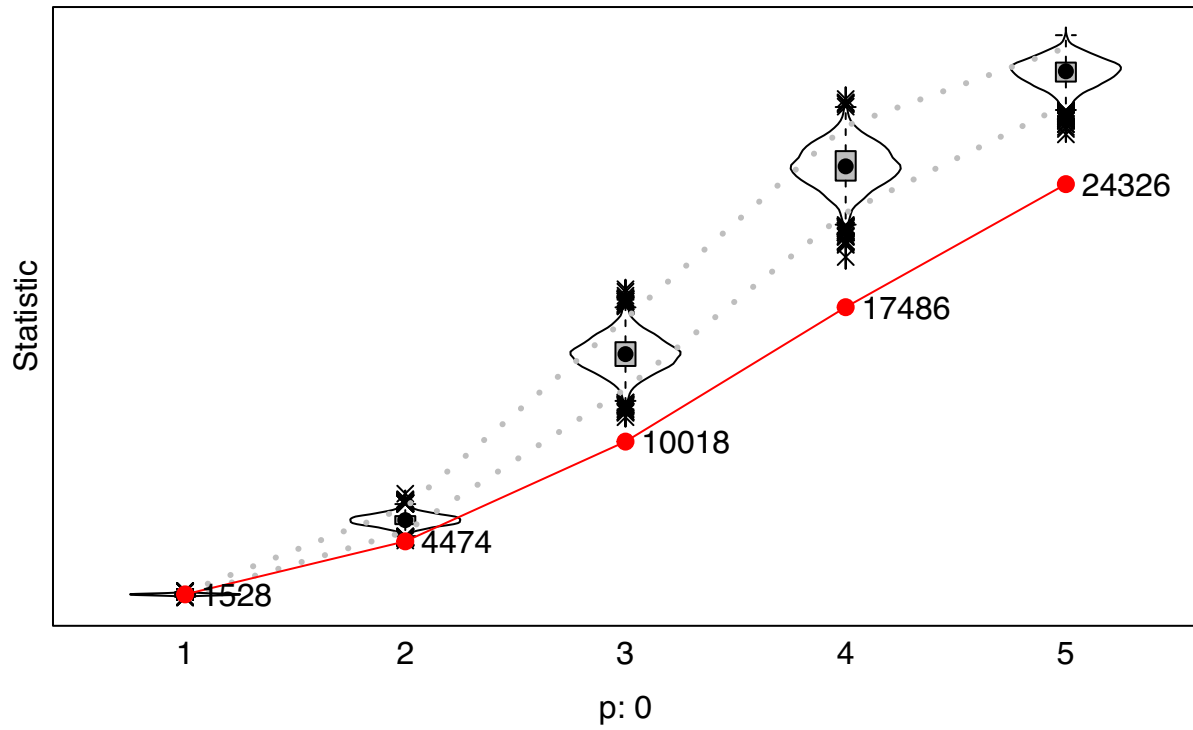
Goodness of Fit of OutdegreeDistribution



```
plot(gofGD)
```

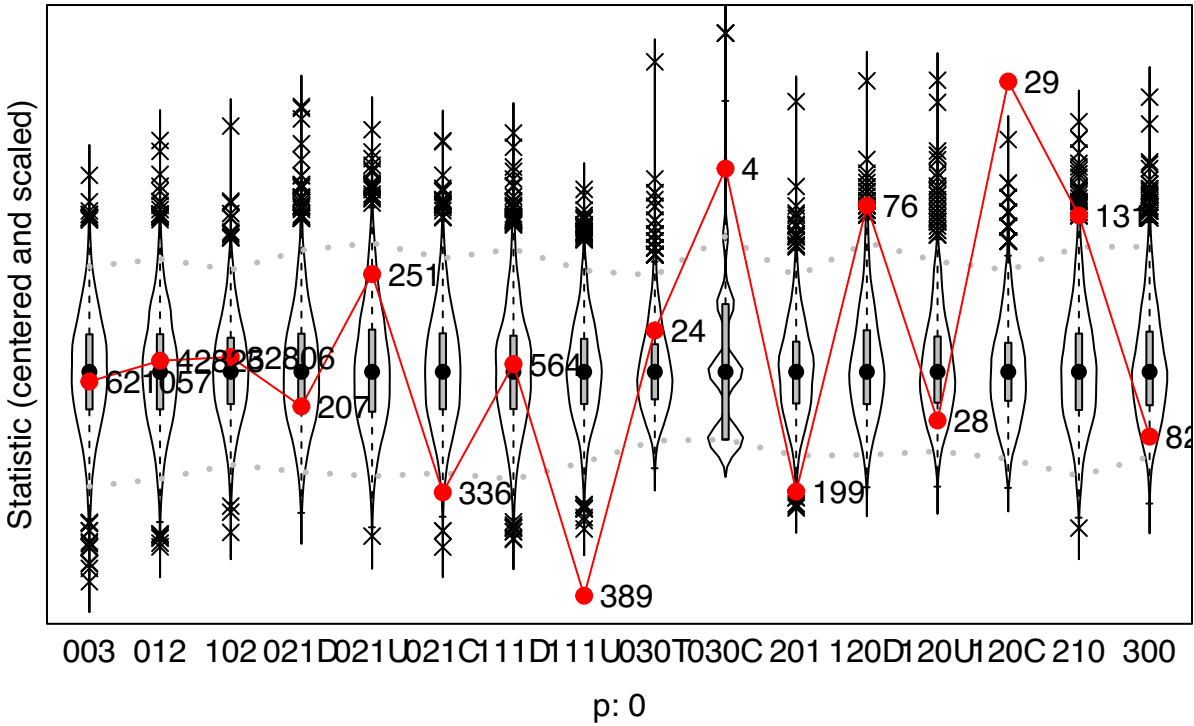
```
## Note: some statistics are not plotted because their variance is 0.  
## This holds for the statistic: Inf.
```

Goodness of Fit of GeodesicDistribution



```
plot(gofTC, center = TRUE, scale = TRUE)
```

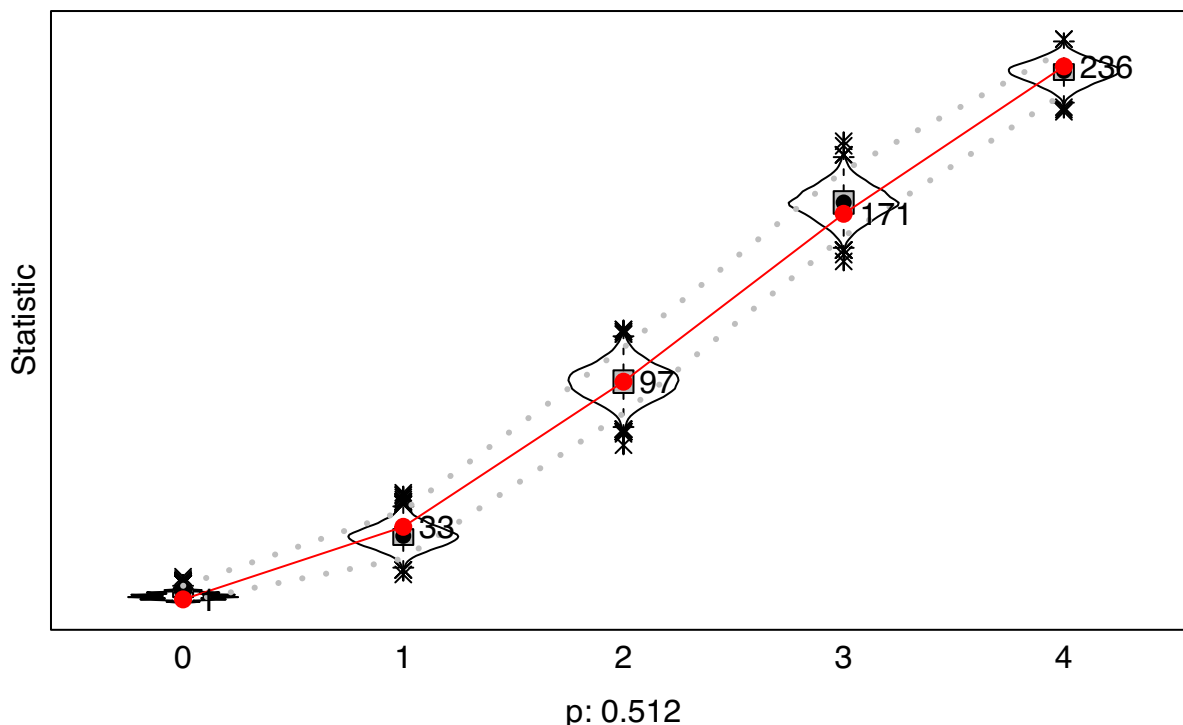
Goodness of Fit of TriadCensus



```
plot(gofBeh)
```

```
## Note: some statistics are not plotted because their variance is 0.
## This holds for the statistic: 5.
```

Goodness of Fit of BehaviorDistribution



Comments: Looking at the plot for the goodness of fit of the indegree distribution, we observe that the red line, i.e. the observed network is almost entirely inside the grey distributions, which are the simulated distributions, across all indegree levels. This shows that the model fits the indegree distribution very well and successfully captures the distribution of the popularity in the network. Also, the high p-value of 0.84 further confirms that there is no significant deviation between the observed distributions and the simulated ones.

In contrast, in the plot for the goodness of fit of the outdegree distribution we see a poor model fit. The observed network (red line) lies mostly outside the range of simulated distributions (gray bands). This suggests that the model doesn't adequately represent the distribution of outgoing ties among students. The p-value of 0 further supports these observations and indicates a significant discrepancy between the observed and simulated distributions and therefore we have strong evidence of model misfit for the outdegree distribution.

In the plot for the goodness of fit of the geodesic distance distribution we see a poor model fit. The observed network (red line) is mostly outside the range of simulated distributions (gray bands) and the observed values are consistently lower than the simulated values. More specifically, the observed network seems to have shorter path lengths between students compared to the simulated one. This suggests that the model doesn't adequately represent the geodesic distance distribution. The p-value of 0 further supports these observations and indicates a significant discrepancy between the observed and simulated distributions and therefore we have strong evidence of model misfit for the geodesic distance distribution.

The plot for goodness of fit of the triad census also shows a poor fit, since several observed triad counts (red line) are outside the range of the simulated distributions (gray bands). The p-value of 0 further supports that the model doesn't accurately represent the triad census distribution and doesn't fully capture all the aspects of the local network structure, even though it includes transitive triplets and three-cycles.

Lastly, the plot for the goodness of fit for the behavior distribution shows that the observed values (red line) lie within the range of the simulated distributions (gray bands) across all levels of alcohol consumption.

This shows that the model successfully captures the distribution of alcohol consumption among students in the network. Also, the high p-value of 0.523 further confirms that there is no significant deviation between the observed distributions and the simulated ones. Therefore, the model provides a very good fit for the behavioural component.

Overall, we see that although the model successfully captures the distribution of popularity and behaviour, it fails to reproduce the outdegree, the geodesic distance and the triad census distributions. This suggests that additional effects may be required to capture the full structure of the network. Comparing these results to the goodness of fit of the friendship-only model from 3.1.4, the co-evolution model performs similarly overall. The indegree fit improves slightly, however the outdegree, geodesic distance and triad census remain poorly fitted in both models, indicating that these are persistent structural issues.

Task 3.2.4

To evaluate the hypotheses iv and v, we examine the effects of alcohol consumption in the co-evolution model. For Hypothesis iv, we were testing whether popular students tend to increase maintain their level of alcohol consumption. To do this, we look at the estimate for the indegree effect on alcohol behaviour. From our results, the estimated coefficient was positive (0.07), but it was not statistically significant (the standard error was 0.07). This means that there is no clear evidence to support that popularity increases alcohol consumption and therefore hypothesis iv is not supported by the data.

For Hypothesis v, we were testing whether students tend to adjust their alcohol consumption to that of their friends. To do this, we look at the average alter effect. The estimated coefficient is positive (1.03) and statistically significant (with a standard error of 0.40). This indicates that students tend to adjust their alcohol consumption levels towards the average alcohol consumption level of their friends. This means that we have strong evidence of students influencing each other in alcohol consumption, and therefore hypothesis v is supported by the data.

Comparing the results above with the results from hypotheses i-iii from Task 3.1, we observe that the conclusions are quite similar between the two models. In both the selection-only model and the co-evolution model, the indegree popularity effect is negative and statistically significant, indicating that students do not tend to form friendships with popular pupils. Similarly, the alcohol similarity effect remains positive and significant, suggesting that students tend to form ties with students that have similar alcohol consumption levels to them. The distance effect also remains negative and significant, showing that students tend to form friendships with students that live closer to them.

However, these effects do differ slightly in magnitude between the two models. For example, when we look at the alcohol similarity effect, it is stronger in the co-evolution model. This indicates that when behavioural dynamics are accounted for, similarity in alcohol consumption is more important in forming ties with other students than in the selection-only model. The key difference between the two models is that in the selection-only model, although we could observe a significant effect of similarities in alcohol consumption on the formation of friendships between students, we were not able to distinguish whether the similarity in alcohol consumption reflected a tendency to select similar friends or a tendency for friends to become more similar over time. With the co-evolution model, we could make that distinction and from the results, we observed that students both choose friends with similar alcohol consumption, but also adjust their behaviour in alcohol consumption to match that of their friends.

Overall, the results of the models suggest that homophily and social influence both play important roles in forming friendships, while popularity does not appear to drive either friendship formation or alcohol consumption.

Task 3.2.5

From the estimates in the co-evolution model, we find evidence for both selection and influence processes in the evolution of friendships and alcohol consumption.

Regarding selection processes, we observed that the alcohol similarity effect is positive and statistically significant, indicating that students tend to form ties with pupils that have similar alcohol consumption levels to them. Also, the geographical proximity and gender effects were also shown to be significant. This suggests that homophily plays an important role in the formation of friendships.

Additionally, regarding influence processes, the alter effect on alcohol consumption is positive and statistically significant, indicating that students tend to adjust their alcohol consumption levels to that of their friends. This means that once friendships are formed, students tend to be influenced by their friends and adjust their behaviour to match that of their friends.

As we saw, there was no evidence that popularity affects alcohol consumption, as the indegree effect on alcohol behaviour was not statistically significant. However, the indegree effect in the friendship network is negative and significant, indicating that students are less likely to form ties with already popular peers. This suggests that popularity does not play an important role in behavioural influence or positive selection in this network. Influence operates mainly through students' similarities, rather than their popularity status in the network.

Overall, the results show that we have evidence for both selection and influence processes, as students tended to form friendships with students similar to them, but they also tended to adjust their behaviour to match that of their friends. This shows that it is important that we consider both selection and influence processes simultaneously, because if we do not, then we may fail to reach correct conclusions about the role of similarity and social influence in the network and behavioural dynamics.

Task 3.3

From the previous parts of this task, the goodness of fit results for the co-evolution model showed that the model reproduced well the indegree and alcohol behaviour distributions. However, the model fit for the outdegree distribution, the geodesic distances and local structure configurations such as triad census, was poor. Therefore, the model could be improved by adding new effects so that geodesic distances and degree distributions might be better represented.

The poor model fit for the outdegree distribution suggests that some students are much more socially active than others and the current model doesn't account for this variation. Therefore, to improve the fit of the degree distribution, and specifically the outdegree distribution, we could include an outdegree activity effect. Some students are naturally more active in seeking friendships than others, and the current model does not account for this variation.

Also, we could include effects such as outdegree popularity and indegree activity, which would help the model capture the dependence between students' activity and popularity better. These effects could help capture the differences between the number of friendship nominations that students send, but also help represent how popularity influences the likelihood of receiving ties, which may not be fully captured just by the indegree popularity effect.

The poor model fit of the geodesic distances distribution suggests that the observed network is more tightly connected than the simulated ones. In order to improve the representation of geodesic distances, we could add effects that capture broader network connectivity. This could be covariates such as students sharing classes or extracurricular activities, since students that interact more frequently within shared contexts are likely to form ties and reduce the distance between them. Since the model does not include those covariates it underestimates the level of connectivity observed in the real network.

Task 3.4

To further investigate the dynamics of friendship and alcohol consumption using SAOMs we could look at the following 2 hypotheses:

Hypothesis 1: Higher alcohol consumption levels increase the likelihood of receiving friendship nominations. (Note that this hypothesis is distinct from hypothesis iv, which tests the opposite causal direction - whether popularity drives alcohol consumption). This hypothesis is based on the idea that engaging in certain behaviours such as alcohol consumption could increase social attractiveness and help in being included in social groups. To test this hypothesis, the alcohol consumption effect should be included as an alter effect in the friendship evaluation function in the following way:

```
includeEffects(myeff, altX, interaction1 = "alcoholDep", name = "friendship")
```

Then after estimating the model, a positive and statistically significant coefficient would indicate that students with higher alcohol consumption are more likely to be more popular. This effect could improve the model fit by capturing behavioural mechanisms influencing popularity and hence help the model better represent the indegree distribution.

Hypothesis 2: Students that attend the same classes are more likely to form friendships. This hypothesis is based on the idea that students who interact more often and happen to be in the same environments are more likely to communicate, develop social relationships and form friendship ties among them. For this hypothesis, we would have to collect data on whether two students attend a class together or not and construct a binary dyadic covariate which would indicate whether two students share at least one class (1=share at least one class, 0=otherwise). This could then be included in the friendship evaluation function as a dyadic covariate effect in the following way:

```
includeEffects(myeff, X, interaction1 = "sameClass", name = "friendship")
```

Then after estimating the model, a positive and statistically significant coefficient would indicate that sharing classes increases the likelihood of students forming a friendship tie. This effect could improve the model fit by capturing shared social contexts and hence improve the representation of geodesic distances.